WSQ Course Code: TGS-2024043854

Build a Human–AI Workforce with Autonomous AI Agents

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Tertiary Infotech Academy Pte Ltd
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HOUSEKEEPING

Digital Attendance



Attendance is mandatory for SSG funding.

Take AM, PM and Assessment digital attendance for every session.

Scan the SSG digital attendance QR code shown by your trainer with your phone camera.

The TRAQOM survey and Assessment attendance are taken on assessment day.

Minimum

75%

attendance required

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About the Trainer

?

[Trainer Name]

[Title / Role]

Qualifications: _____

Expertise: _____

Experience: _____

Contact / Profile: _____

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About the Trainer

AA

Dr. Alfred Ang

AI Trainer & Engineer
Tertiary Infotech Academy

Qualifications PhD-qualified trainer specialising in AI, automation and software engineering.

Expertise Builds autonomous AI agents and multi-agent workflows with Hermes Agent, Paperclip and Claude Code.

Experience Trains professionals across Singapore on practical, project-based AI skills.

Certifications PMP, AWS, Microsoft, CompTIA, CKA, CKAD.

Profile linkedin.com/in/angchewhoe · github.com/alfredang

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INTRODUCTION

Let's Know Each Other

Take a minute to introduce yourself to the class:



Your name & role



What you do day to day



Your experience with AI or agents



One task you want an AI agent to take off your plate

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HOUSEKEEPING

Ground Rules

Phones on silent

Step out quietly for calls or breaks.

Participate actively

No question is too small — ask freely.

Mutual respect

Agree to disagree; one conversation at a time.

Be punctual

Return from breaks on time.

75% attendance

Required for funding eligibility.

Stay hands-on

Follow along with every lab.

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ROADMAP

Course Outline

Three topics, ten connected use-case labs, one company: Kopi Kaki Co..

1

Foundations of Autonomous AI Agents & the Human–AI Workforce

- A brief history of AI
- AI agents vs agentic AI
- The agentic loop & harness engineering
- Brain · perception · action, memory & world models
- From single agent to human–AI collaboration
- The 2026 agent market, use cases & governance

2

Hermes Agent — From a Single AI Employee to a Multi-Agent Team

- Lab 1 · Your first AI employee
- Lab 2 · End-to-end social media campaign
- Lab 3 · Human-in-the-loop publishing
- Lab 4 · Multi-agent video studio
- Lab 5 · Campaign command centre

3

Paperclip — Running & Governing a Company of AI Agents

- Lab 6 · Found the AI company
- Lab 7 · Hire the team
- Lab 8 · Launch campaign project
- Lab 9 · Weekly video routine
- Lab 10 · Govern, measure & improve

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WHAT YOU'LL ACHIEVE

Learning Outcomes

Mapped to the TSC Artificial Intelligence Application in Product Development (ICT-TEM-4034-1.1).



LO1 · Analyse AI applications

Analyse AI (LLM-based) applications across a range of industries to identify their capabilities and limitations.



LO2 · Agent design and efficiency

Establish the relationship between AI algorithm/agent design and chatbot/agent efficiency.



LO3 · Evaluate & improve

Evaluate and improve the effectiveness of AI (RAG and agent) applications in product development.

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WHAT YOU'LL ACHIEVE

Learning Units, Topics and Labs

Approved Course Proposal: TSC ICT-TEM-4034-1.1 Artificial Intelligence Application in Product Development — 16 hours.

LU	Learning unit	Delivered through	K & A	Hours
LU1	Foundations of autonomous AI agents and single-agent LLM applications	Topic 1 (Foundations) + Topic 2 Labs 1–2 (single Hermes agent)	K1, K6, A1, A3	5 h
LU2	Agent and multi-agent design for efficiency, with humans in the loop	Topic 2 Labs 3–5 (human-in-the-loop, multi-agent studio, swarm) + Topic 3 Lab 6 (designing a governed agent company)	K4, K5, A2, A6	5 h
LU3	Evaluate and improve the effectiveness of an AI-agent workforce in product development	Topic 3 Labs 7–10 (team, launch project, routine, governance and improvement)	K2, K3, A4, A5	4 h
Assess	WA-SAQ (1 h) + Practical Performance (1 h)	Day 2, 4:30–6:30 PM	K1–K6 · A1–A6	2 h

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ROADMAP

Lesson Plan — Day 1 (9:30 AM – 6:30 PM)

Time	Session	LU
9:30–9:50	Welcome, housekeeping, trainer introduction, learning outcomes and AM digital attendance	LU1
9:50–11:05	Topic 1 (Part A) — history of AI; AI agents vs agentic AI; the agentic loop; harness engineering; brain–perception–action; short- and long-term memory; RAG; world models	LU1
11:05–11:20	Tea break	LU1
11:20–12:10	Topic 1 (Part B) — single agent → multi-agent → human–agent collaboration; levels of autonomy; agent society; the 2026 agent market (OpenClaw, Hermes, Paperclip, OpenAI Dots, Meta Muse, Block Buzz, Manus Cue …); Singapore's AgentSea; human–AI use cases in science, maths, drug discovery and business; risks and governance	LU1
12:10–12:35	Activity — Human–AI Workforce Canvas for Kopi Kaki Co., and start the Hermes Agent install	LU1
12:35–13:30	Topic 2 — Lab 1: Hire Your First AI Employee (Hermes) — customer & market research brief	LU1
13:30–14:30	Lunch break	—
14:30–15:30	Lab 2: End-to-End Social Media Campaign with one agent (skills + tools)	LU1
15:30–16:15	Lab 3: Human-in-the-Loop Publishing & Community Care (Telegram + cron)	LU2
16:15–16:30	Tea break	LU2
16:30–17:30	Lab 4: Multi-Agent Video Studio — research → script → production → audit → publish	LU2
17:30–18:15	Lab 5: Multi-Agent Campaign Command Centre — measure, optimise & report	LU2
18:15–18:30	Day 1 recap, Q&A and PM digital attendance	LU2

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ROADMAP

Lesson Plan — Day 2 (9:30 AM – 6:30 PM)

Time	Session	LU
9:30–9:45	Day 1 recap and AM digital attendance	LU2
9:45–10:15	Topic 3 — Paperclip: the company as the agent harness; Board, CEO and org chart; adapters; goals, projects and tasks; routines and heartbeats; budgets, approvals and the audit trail	LU2
10:15–11:15	Lab 6: Found Kopi Kaki Growth Studio — an AI company you govern	LU2
11:15–11:30	Tea break	LU2
11:30–12:15	Lab 7: Hire the Marketing Team through the Board approval gate	LU3
12:15–13:15	Lunch break	—
13:15–14:10	Lab 8: Run the #ShiokColdBrew launch as a project — review, revise, release	LU3
14:10–14:55	Lab 9: Automate the weekly video pipeline with a routine and a Board gate	LU3
14:55–15:10	Tea break	LU3
15:10–15:55	Lab 10: Govern, measure & improve your human–AI workforce	LU3
15:55–16:15	Course recap, practice exam and Q&A	LU3
16:15–16:30	Briefing for Assessment	LU3
16:30–17:30	Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book	Assess
17:30–18:30	Practical Performance (PP) — 1 hour, open book; TRAQOM survey and PM / assessment digital attendance	Assess

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ROADMAP · HANDS-ON

Ten Labs, One Company

Every lab builds on the last — the same Kopi Kaki campaign grows from one agent to a governed AI company.

LAB 1 · DAY 1

First AI Employee

Single agent

LAB 2 · DAY 1

Social Media Campaign

Single agent + skills

LAB 3 · DAY 1

Human-in-the-Loop

Human–agent collaboration

LAB 4 · DAY 1

Multi-Agent Video Studio

Multi-agent (ordered pipeline)

LAB 5 · DAY 1

Campaign Command Centre

Multi-agent (parallel swarm)

LAB 6 · DAY 2

Found the AI Company

Human-governed company of agents

LAB 7 · DAY 2

Hire the Team

Multi-agent organisation

LAB 8 · DAY 2

Launch Campaign Project

Multi-agent project with Board review

LAB 9 · DAY 2

Weekly Video Routine

Scheduled multi-agent workflow

LAB 10 · DAY 2

Govern & Improve

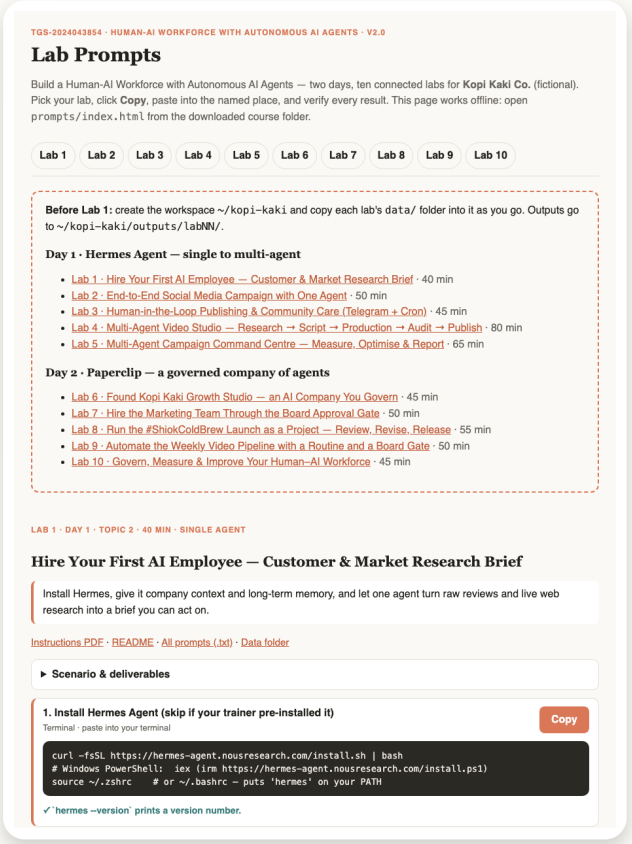
Workforce governance

TIP Day 1 = Hermes Agent (single → multi-agent). Day 2 = Paperclip (a governed company of agents).

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Lab Materials & the Lab Prompts Page

Everything you need is in the course repo: instructions PDF, mock data, skills and copy-ready prompts for every lab.



Get the labs

github.com/tertiarycourses/TGS-2024043854-… → Code → Download ZIP. Unzip anywhere.

Open the prompts page

Double-click prompts/index.html — pick your lab, click Copy, paste into Hermes, Telegram or Paperclip.

Use the lab folder

labs/labNN-…/ holds README, LAB-NN-Instructions.pdf, COPY-PROMPTS.txt and data/.

Verify every result

Each lab ends with a checkpoint — evidence, not vibes.

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THE COURSE SCENARIO

Meet Kopi Kaki Co.

A fictional Singapore kopi chain — the company every lab works for. All data is synthetic.



12 outlets · 140 staff

Heritage kopi, modern craft — Tiong Bahru to Changi T3.



Launch: Gula Melaka Cold Brew

SGD 6.80 · 1 Dec 2026 · code SHIOK20 (20% off first app order, 1–14 Dec).



Campaign #ShiokColdBrew

17 Nov – 14 Dec · SGD 15,000 · 2,500 pre-orders · 8% engagement · 40k views.



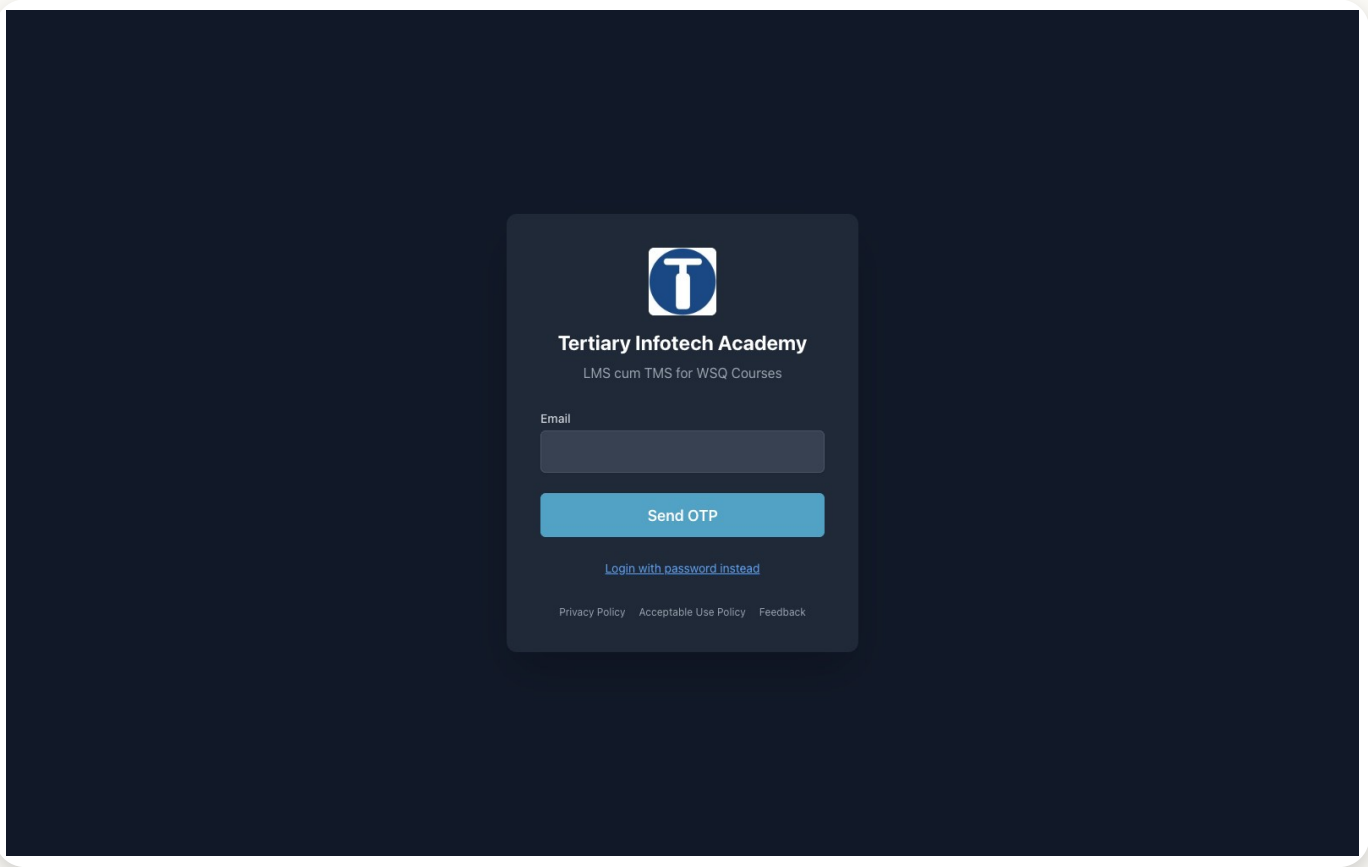
Rules that matter

No health or halal claims · no competitor bashing · human APPROVE before anything is public.

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LEARNING MANAGEMENT SYSTEM

Courseware & Assessment on the LMS



Log in

lms-tms.tertiaryinfotech.com with your registered email + OTP.

Download the courseware

Slides, Learner Guide and Lesson Plan for the open-book assessment.

Take the assessment

Complete the WA and PP and upload your answers on the LMS.

- LMS: https://lms-tms.tertiaryinfotech.com/ → open this course → Courseware tab.

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LEARNING MANAGEMENT SYSTEM

Download Course Material

STEP 1



Open the LMS

Go to lms-tms.tertiaryinfotech.com

STEP 2



Send OTP

Enter your registered email and click Send OTP.

STEP 3



Verify

Key in the one-time password from your inbox.

STEP 4



Open the course

Select Build a Human-AI Workforce with Autonomous AI Agents.

STEP 5



Download

Slides PDF + Learner Guide — keep them for the open-book assessment.

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ASSESSMENT

Briefing for Assessment

Clear your space

Phones and other materials under the table or on the floor.

No recording

No photos or recording of assessment scripts.

Open book

Slides, Learner Guide and approved materials only.

Work individually

No discussion during the assessment.

Manage your time

WA (1 hour) then PP (1 hour).

Appeals

An appeal process is available if you disagree with the outcome.

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ASSESSMENT

Assessment & Funding

Final Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — three tasks, each with ONE screenshot from a lab (Lab 1 Hermes agent, Lab 4 multi-agent studio, Lab 7 Paperclip org chart) plus written analysis, 1 hour, open book.
- Day 2, from 4:30 PM — open book
- Allowed: slides, Learner Guide, approved materials
- Appeal process available

Criteria for Funding

Minimum 75% attendance
Based on SSG digital attendance records.

Assessed as 'Competent'
Complete and pass both assessment components (WA and PP).

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ASSESSMENT

Assessment Flow

On assessment day, follow these five steps in order.

STEP 1



TRAQOM

Scan the TRAQOM QR code on the LMS and complete the survey.

STEP 2



Attendance

Take the Assessment digital attendance (SSG QR).

STEP 3



Assessment

Sit the WA (SAQ), then the PP — open book.

STEP 4



Submit

Upload your answers on the LMS.

STEP 5



Sign the record

Sign the Assessment Summary Record.

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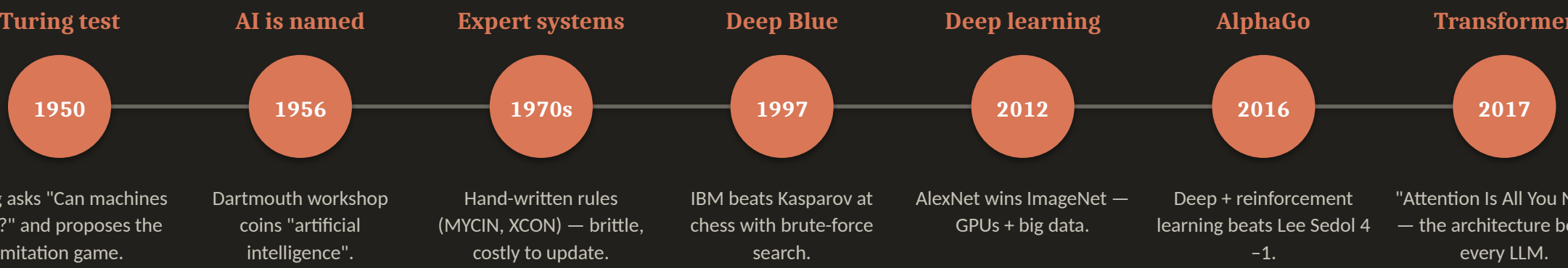
TOPIC 1

Foundations of Autonomous AI Agents & the Human–AI Workforce

1. A brief history of AI
2. AI agents vs agentic AI
3. The agentic loop & harness engineering
4. Brain · perception · action, memory & world models
5. From single agent to human–AI collaboration
6. The 2026 agent market, use cases & governance

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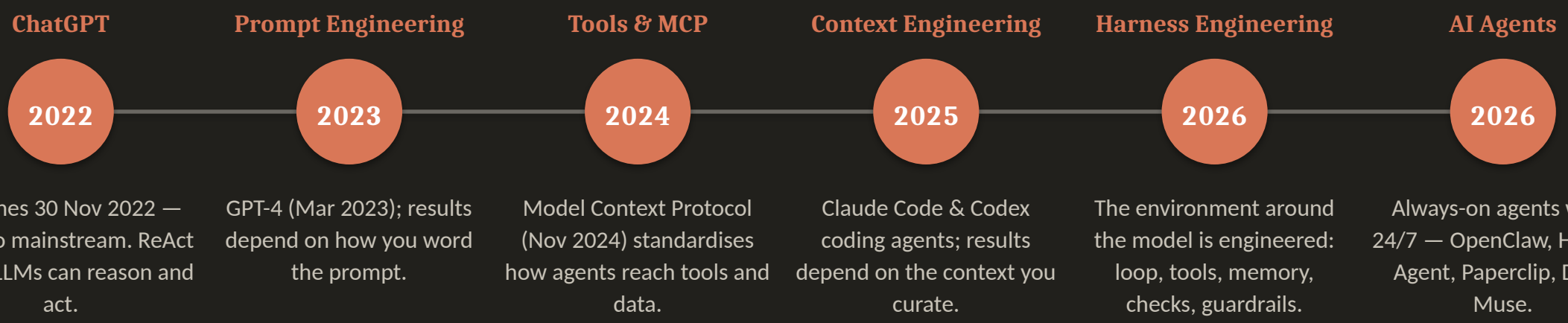
A Brief History of AI — From Rules to Learning



Rules → Search → Learning → Language: each era moved more of the 'how' from humans into the machine.

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A Brief History of AI, 2022–2026 — From Chat to Agents



From wording one prompt → curating context → engineering the harness → agents that work around the clock.

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Prompt, Context & Harness Engineering

Each layer contains the one before it.

Prompt engineering

Craft the instruction. One request, one answer. Quality depends on wording.
Example: "Write a caption for our new drink."

Context engineering

Curate what the model sees: brand guide, brief, data, memory, tool results — and nothing that wastes the window.
Example: AGENTS.md + campaign-brief.md loaded every session.

Harness engineering

Design the whole environment: the loop, tools, memory, skills, verification, guardrails and human gates — so the agent does reliable long-running work.
Example: Lab 4's five-agent studio with an auditor and an approval gate.

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The Range of AI Applications Today

Every industry uses several of these together — agents increasingly orchestrate the others.

Generative AI / LLMs

Chatbots, writing, coding and summarising. Capability: understand and generate natural language, code and images.

Computer vision

Defect detection, medical imaging, product search. Capability: recognise and classify what is in images and video.

Predictive analytics & ML

Demand forecasting, fraud scoring, recommendations. Capability: learn patterns in data to predict outcomes.

Speech & multimodal

Voice assistants, transcription, text-to-speech. Capability: convert between speech, text and other media.

Scientific discovery

AlphaFold, AI co-scientist, AlphaEvolve. Capability: generate and rank hypotheses and designs for experts to test.

Autonomous AI agents

Hermes, Paperclip, OpenAI Dots, AgentSea. Capability: pursue a goal with tools, memory and planning — with humans in the loop.

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What Is an AI Agent?

"Autonomous software entities engineered for goal-directed task execution within bounded digital environments" — Sapkota et al., 2025

Goal-directed

Given a goal, it decides the steps itself instead of waiting for each instruction.

Uses tools

Searches the web, reads files, writes documents, calls APIs and MCP servers.

Loops until done

Acts, observes the result, checks it, and tries again.

Remembers

Keeps context in the session and long-term memory across sessions.

Bounded

Works inside permissions, budgets and approval gates set by humans.

Accountable human

A person owns the outcome — the agent recommends or acts on their behalf.

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AI Agents vs Agentic AI

Source: Sapkota, Roumeliotis & Karkee, "AI Agents vs. Agentic AI", Information Fusion 2025 (arXiv 2505.10468).

Dimension	AI Agent	Agentic AI
Scope	One specific, well-defined task	A complex, multi-step goal or workflow
Architecture	One LLM + tools: perceive → reason → act	Several specialised agents + an orchestrator
Flow	User → Agent → Tool	User → System → Agent 1 → Agent 2 → …
Memory	Session context, optional cache	Shared, persistent memory across stages
Planning	Single-step, reactive	Multi-step, proactive goal decomposition
Coordination	Works alone	Hierarchical or decentralised hand-offs
Examples	Support bot, email triage, research agent (Lab 1)	Video studio (Lab 4), AI company (Labs 6–10)
Main risks	Hallucination, brittle prompts	Error propagation, misalignment, governance

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An Analogy: the Smart Home

AI Agent — a smart thermostat

- Holds the room at 21°C
- One goal, one device, clear rules
- Reacts to the temperature it senses
- Kopi Kaki: Mia writes a research brief (Lab 1)

Agentic AI — the coordinated home

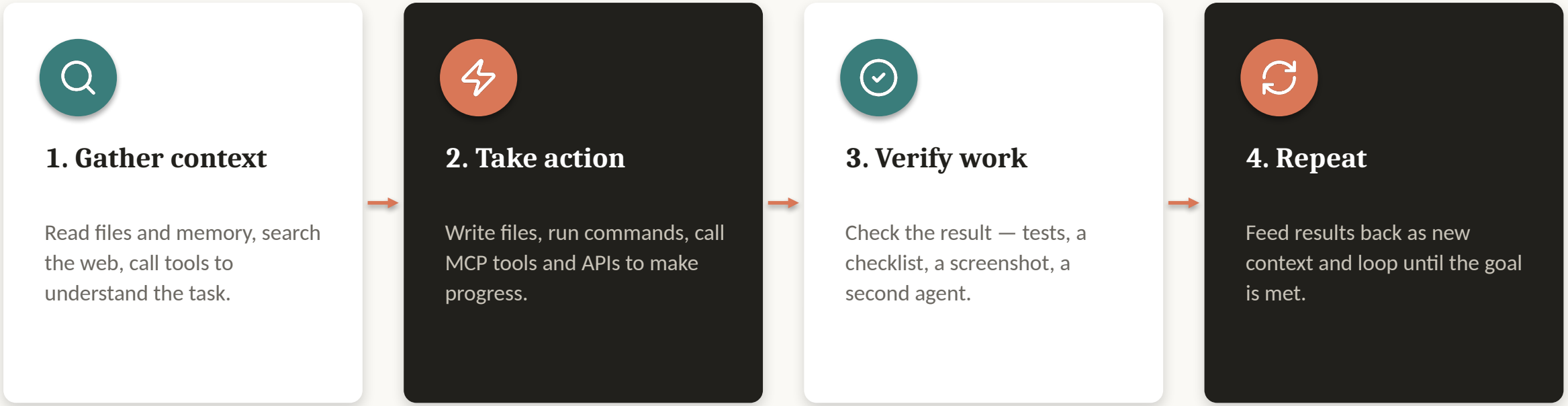
- Weather, energy-price, schedule, appliance and security agents work together
- Pre-cools with solar before a heatwave hits at peak price
- Plans ahead, hands off, adapts the goal
- Kopi Kaki: a 5-agent studio ships a video (Lab 4)

TIP Same LLMs underneath — the difference is orchestration, memory and coordination.

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The Agentic Loop

Every agent task runs the same repeating cycle. ReAct (2022) calls it Thought → Action → Observation.



Goal in → context → action → verification → repeat → goal met

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The Loop in Action — Mia's Research Brief (Lab 1)



Goal

"Write a research brief with three evidence-backed campaign angles."



Gather context

Loads AGENTS.md and memory, reads 60 reviews, searches 2026 cold-brew trends.



Reason & plan

Groups reviews into themes, compares competitors, drafts three angles.



Act

Writes review-insights.md and research-brief.md with citations.



Verify

Checks each claim has a review ID or URL; flags anything unverified.



Human checkpoint

You challenge a claim and choose the angle — Mia saves it to long-term memory.

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Why the Loop Matters

Give direction, not micro-steps

State the goal, constraints and definition of done; let the loop find the steps.

Verification is everything

An agent is only as good as its ability to check its own work — checklists, auditors, tests.

Steer mid-loop

Review plans and drafts early; send work back with specific feedback rather than waiting for a wrong final answer.

Context is the fuel

What the agent can see decides what it can do — that is why we give it AGENTS.md, skills and memory.

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What Is Harness Engineering?

Agent = Model + Harness. The harness is everything around the model that turns it into a dependable worker.

A harness provides

- The agentic loop (gather → act → verify → repeat)
- Tools and MCP connections
- Context & memory management
- Skills (procedural know-how)
- Verification — checklists, tests, auditor agents
- Guardrails — permissions, budgets, approval gates
- Sub-agents and orchestration
- Observability — logs, costs, audit trail

Who named it

Nov 2025 — Anthropic: "Effective harnesses for long-running agents".

Feb 2026 — Mitchell Hashimoto: "Engineer the harness" — every time an agent makes a mistake, fix the environment so it never happens again.

Feb 2026 — OpenAI Codex team: ~1M lines of code, ~1,500 PRs, no hand-written code. "Humans steer. Agents execute."

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The Harness in This Course

Context: AGENTS.md

Company facts and rules loaded every session (Lab 1).

Memory

MEMORY.md / USER.md recall decisions across sessions (Lab 1).

Skills

kopi-brand-voice, campaign planner, compliance audit (Labs 2, 4).

Verification

Self-check tables, auditor agent, verifier in the swarm (Labs 2, 4, 5).

Guardrails

APPROVE on Telegram, blocked publish task, Board gates (Labs 3, 4, 7–9).

Orchestration

Profiles + Kanban, swarms, Paperclip org chart (Labs 4–10).

Scheduling

Cron jobs and Paperclip routines (Labs 3, 5, 9).

Observability

Kanban history, costs and the audit log (Labs 5, 10).

TIP When an agent gets it wrong, improve the harness (instructions, skills, gates) — don't just re-prompt.

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The Agent Framework: Perception · Brain · Action

From the survey "The Rise and Potential of LLM-Based Agents" (Xi et al., Fudan University, arXiv 2309.07864).

Perception

Sense the world

- Multimodal input — text, image, audio, video
- Environment interaction — tools, web, files
- Information retrieval — search, RAG, memory

From data to information

Brain

Core cognition & decisions

- Knowledge — pre-trained, domain, common sense
- Memory — short-term & long-term
- Reasoning — logic, reflection, self-correction
- Planning — decompose, set goals, adapt

From understanding to decisions

Action

Change the world

- Task execution — code, tools, operations
- Environment feedback — results, rewards
- Behaviour optimisation — learn and evolve

From decisions to action

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Inside the Brain

The LLM is the control centre: it turns what the agent perceives into decisions.

Knowledge

Pre-trained and domain knowledge, common sense and self-knowledge. Can be stale → ground it with RAG.

Memory

Short-term (the context window) and long-term (notes, history, skills). Retrieved when relevant.

Reasoning

Logical and step-by-step reasoning, reflection and self-correction on complex problems.

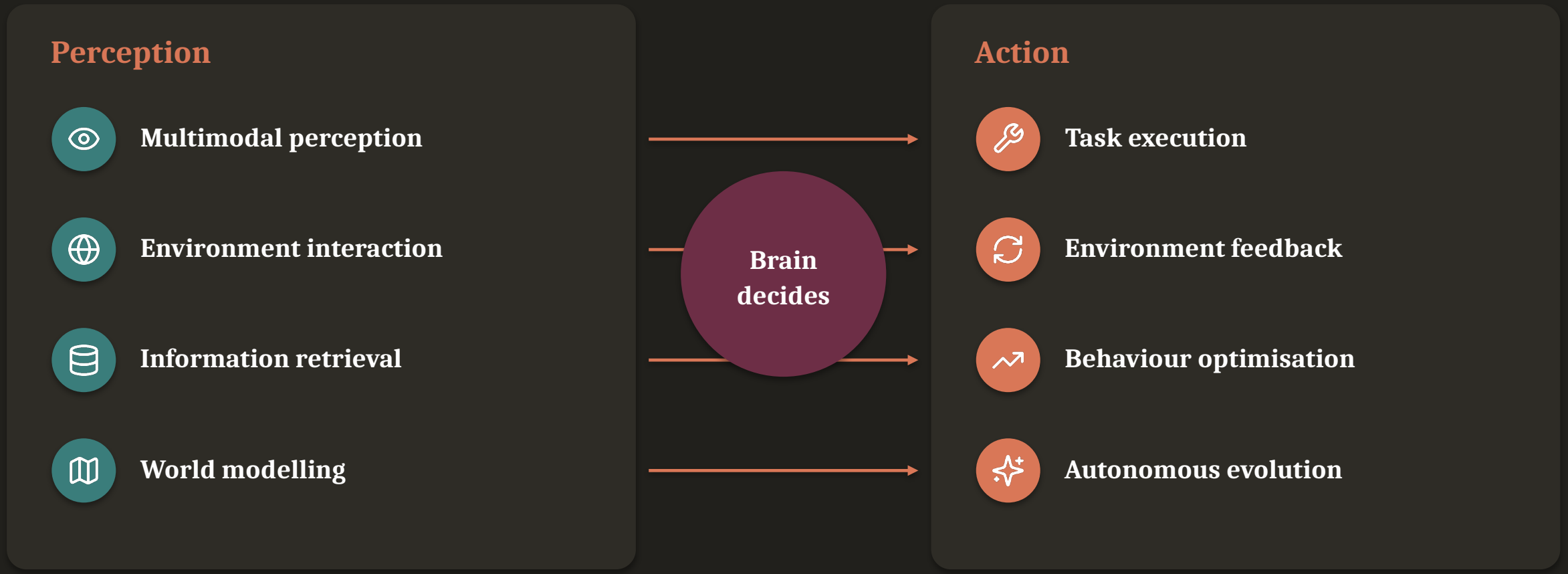
Planning

Break tasks into sub-goals, set targets, revise the plan after feedback, improve over time.

TIP Hermes adds natural-language interaction and transfer: one agent, every channel, new skills learned on the job.

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Perception → Decision → Action: a Closed Loop



Perception → decision → action → feedback → new perception. In Lab 3 the feedback is your Telegram reply.

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Memory: Short-Term and Long-Term

Framework: CoALA (Sumers et al., 2023). Short-term memory is limited and resets; long-term memory persists and is retrieved when relevant.

Memory type	Human analogy	What the agent keeps	Where you will see it
Short-term / working	What you're thinking about now	The context window: instructions, chat, tool results, current plan	Every Hermes session; compaction when it fills
Episodic (long-term)	"What happened"	Past sessions, actions and outcomes	Hermes session search; Kanban & Paperclip history
Semantic (long-term)	"Facts I know"	Facts about the company and user; knowledge bases (RAG)	MEMORY.md / USER.md; brand guide in company knowledge
Procedural (long-term)	"How to do things"	Skills, rules, workflows	SKILL.md skills; AGENTS.md; agent instructions

TIP Hermes loads memory at session start — after it saves something, start a new session (/new) to see it recalled.

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How Agents Remember — Retrieval & Reflection

Retrieve by relevance

Pull the memories that score highest for recency, importance and relevance (Generative Agents, 2023).

RAG

Retrieve from a curated, current knowledge base before answering — grounded, auditable answers.

Just-in-time context

Keep pointers (file paths, links) and load the data only when needed — saves the window.

Reflection

After a failure, write a lesson to memory and do better next time (Reflexion, 2023).

TIP Lab 2 challenge: turn your review feedback into a permanent skill update — reflection by design.

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RAG Agent vs a Standard LLM Assistant

RAG agent

- Retrieves from a curated, current corpus before generating
- Grounded answers with citations — less hallucination
- Quality depends on corpus freshness, chunking and retrieval/memory configuration
- Needs a maintenance process: re-index, review sources

Standard LLM assistant

- Answers from training data alone — fastest to deploy
- No live company data; knowledge can be stale
- Harder to audit where an answer came from
- Lower maintenance, higher risk on facts

TIP Evaluate both on accuracy, relevance, latency, cost and maintenance effort.

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World Models — Predicting the Next State of the World

LLMs predict the next word; world models predict how the world changes after an action — so an agent can plan before it acts.

Why they matter

- Plan and simulate "what if I do X?" before costly or irreversible actions
- Physical AI — robots and vehicles need intuitive physics
- Train agents safely in generated worlds
- Ground the agent in how the world really behaves

NVIDIA Cosmos	Jan 2025	World foundation models for robots and autonomous vehicles.
Meta V-JEPA 2	Jun 2025	Video world model; zero-shot robot pick-and-place.
DeepMind Genie 3	Aug 2025	Interactive worlds from text at 24 fps, 720p.
World Labs Marble	Nov 2025	Persistent, editable 3D worlds from text or images.
AMI Labs (LeCun)	Mar 2026	US$1.03B raised to build JEPA-based world models.

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Three Paradigms: Single Agent → Multi-Agent → Human–Agent Collaboration

Xi et al. (2023): from individual agents to collaboration and agent society.

Single agent

One agent completes a task end to end.

- Task-oriented — follows instructions
- Innovation-oriented — research & discovery
- Lifecycle-oriented — learns skills over time

Labs 1–2: Mia researches and runs a campaign.

Multi-agent

Several agents collaborate or compete.

- Cooperative — ordered (SOP pipeline) or disordered (discussion, voting)
- Adversarial — debate to improve answers

Labs 4–5: video studio pipeline and analysis swarm.

Human–agent collaboration

People and agents share the work.

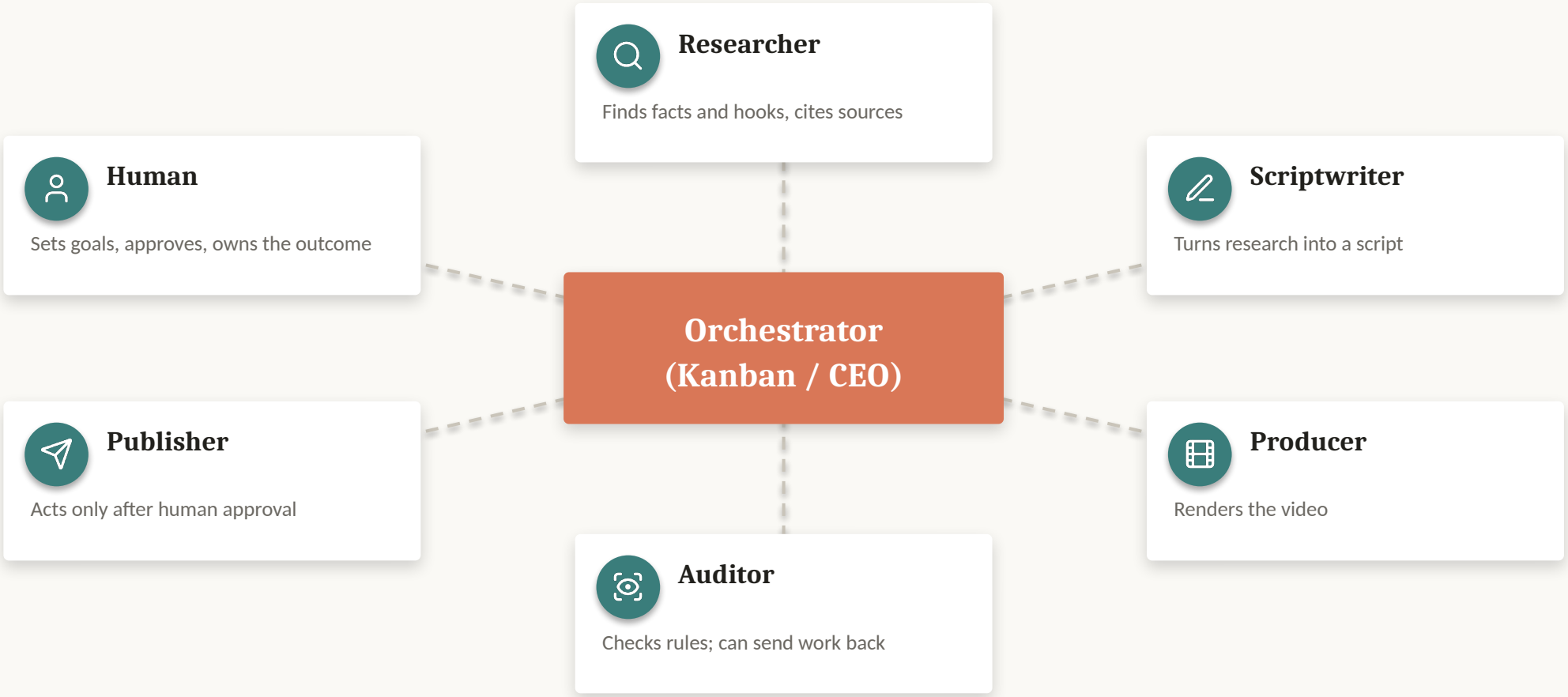
- Instructor–executor — humans instruct and give feedback
- Equal partnership — agents as peers

Lab 3 and Day 2: APPROVE, review and Board gates.

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Multi-Agent Patterns You Will Build

Ordered pipeline (Lab 4), parallel swarm with a verifier (Lab 5), and an org chart with a Board (Labs 6–10).



TIP Without rules, multi-agent chatter can amplify small hallucinations — add verifiers and human checkpoints.

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Two Ways Humans and Agents Work Together

Instructor–executor

- The human sets goals, gives instructions and feedback
- Feedback can be quantitative (scores) or qualitative (critique)
- Agent executes and improves from feedback
- Labs: APPROVE / EDIT on Telegram; Board revision requests

Equal partnership

- Agent participates as a peer — proposes, negotiates, challenges
- Empathetic communicator or human-level participant
- Human remains accountable for decisions
- Labs: the CEO agent proposes the team; you approve or push back

TIP Why keep humans in? They fill gaps in privacy, ethics, judgement and data — and agents learn through interaction.

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Levels of Autonomy — Pick Per Task, Not Per Agent

Feng, McDonald & Zhang (2025): autonomy is a design decision, separate from capability.

Level	Your role	What the agent does	Example in this course
L1	Operator	Acts only when you invoke it for a subtask	Asking Mia a one-off question
L2	Collaborator	Plans and works with you; you can take over anytime	Lab 2 campaign drafting with your feedback
L3	Consultant	Does most of the work; asks for your expertise	Lab 1 research brief, Lab 7 hiring plan
L4	Approver	Asks only at high-risk steps or required approvals	Lab 3 APPROVE gate, Lab 9 weekly routine
L5	Observer	Works on its own; you monitor with an off-switch	Weekly report cron (Lab 5)

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Human-in-the-Loop vs Human-on-the-Loop

Pattern	Definition	Use it for	Course example
Human-in-the-loop	A person approves BEFORE the action happens	Irreversible, public, regulated actions — publishing, payments, refunds	Telegram APPROVE (Lab 3), blocked publish task (Lab 4), Board gates
Human-on-the-loop	The agent acts; a person monitors and can intervene	High-volume, reversible work — triage, drafts, analysis	Weekly report cron (Lab 5), Paperclip costs & activity (Lab 10)
Human-out-of-the-loop	Fully autonomous, no real-time oversight	Only low-risk, bounded, reversible tasks	Scheduled data refresh

TIP Keep humans IN the loop for irreversible actions and ON the loop for reversible volume. The human is always accountable.

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From Individuals to an Agent Society

Individual behaviour

Each agent learns and adapts — Hermes writes new skills after hard tasks.

Group interaction

Agents communicate and coordinate — Kanban hand-offs, swarms, CEO delegation.

Personality

Roles and values in SOUL.md and instructions shape how each agent behaves.

Social emergence

Collective intelligence and new behaviours appear at scale — plus new risks (groupthink, bias).

Agent-as-a-Service

Agents offered as a workforce you hire and govern — the idea behind Paperclip.

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Autonomous Agents in the Market (Oct 2026) — 1/2

Product	Vendor · launch	Type	Runs where	Human-in-the-loop
OpenClaw	OpenClaw Foundation · Nov 2025 (renamed Jan 2026)	Open source personal agent	Your machine; chat apps	DM pairing; optional sandbox
Hermes Agent	Nous Research · ~Feb 2026	Open source, self-improving agent	Local, VPS, desktop; 15+ channels	Approval modes; allow-lists
Paperclip	Paperclip Labs · Mar 2026	Open source company of agents	Self-hosted (Node / Docker)	Board approvals; budgets auto-pause
OpenAI Dots	OpenAI · 29 Sep 2026	Always-on agent with its own cloud computer	ChatGPT, Slack, Teams	Custom rules: allow / approve / block
Meta Muse	Meta · 8 Sep 2026 (US)	Personal agent for life admin	App, web, WhatsApp	Confirms sensitive actions; Sentinel guard
Block Buzz	Block · 21 Jul 2026	Open workspace for humans + agents	Self-host or buzz.xyz	Agent identities & permissions
Manus Cue	Manus · 28 Sep 2026	Agents with email, phone, wallet	Web, desktop, mobile	Budgets; you make the final call

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Autonomous Agents in the Market (Oct 2026) — 2/2

Product	Vendor · launch	Type	Runs where	Human-in-the-loop
Claude Cowork / Claude Code	Anthropic · 2025–2026	Desktop & coding agents with sub-agents	Desktop, terminal, cloud	Manual / Auto / Skip permission modes
OpenAI Codex	OpenAI · May 2025; Cloud Sep 2026	Parallel coding agents	CLI, IDE, cloud	Review before merge; sandboxes
Gemini Enterprise agents	Google · 8 Oct 2026	Sub-agents + persistent 'coworker' agents	Workspace, M365, Slack	Agent identity, gateway, spend caps
Microsoft Agent 365	Microsoft · May 2026	Governance control plane for many agents	M365 admin, Entra, Purview	Registry, access control, DLP
Perplexity Computer	Perplexity · Feb 2026	Multi-model orchestrator	Cloud	Not confirmed

TIP Sources: vendor announcements and repos (verified Oct 2026). Products move fast — check before you buy.

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How to Read the Market

Personal agents

One agent working for one person — OpenClaw, Hermes Agent, Meta Muse, OpenAI Dots, Manus Cue.

Workspaces & companies

Many agents and humans with roles — Paperclip, Block Buzz, Gemini coworker agents.

Governance layers

Registries, identities, budgets and policy across agents — Microsoft Agent 365, Paperclip Board.

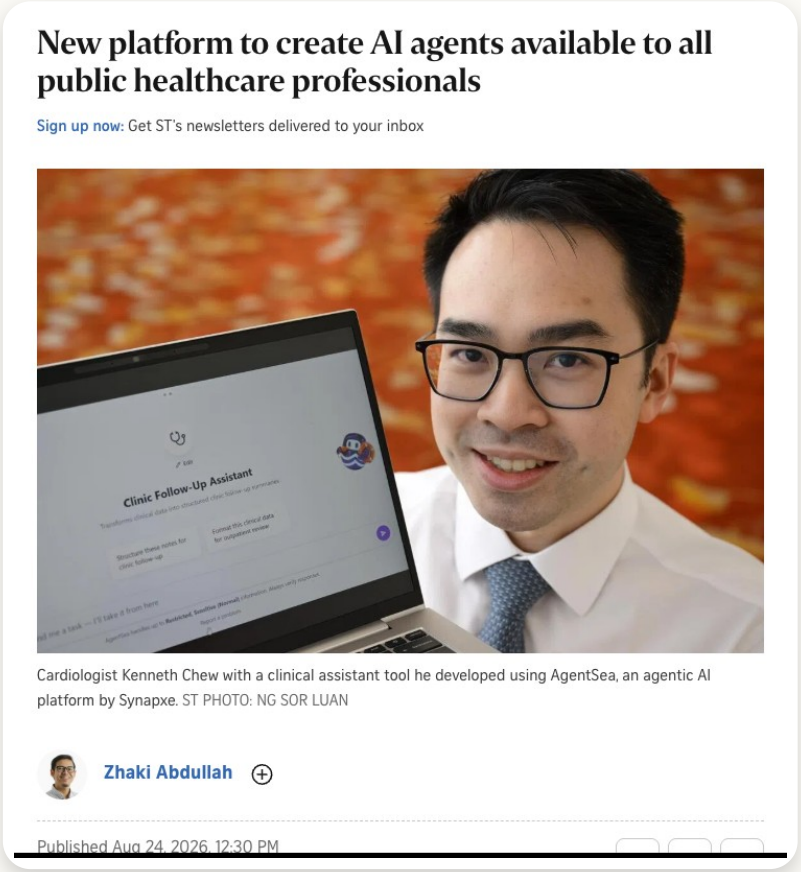
Open vs closed

Open source = self-host, inspect, own your data. Closed = convenience, managed security, vendor lock-in.

TIP This course: Hermes Agent (an employee that learns) + Paperclip (the company you govern) — both open source.

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Singapore: 12,000+ AI Agents in Public Healthcare



New platform to create AI agents available to all public healthcare professionals

Sign up now: Get ST's newsletters delivered to your inbox

Clinic Follow-Up Assistant

Cardiologist Kenneth Chew with a clinical assistant tool he developed using AgentSea, an agentic AI platform by Synapxe. ST PHOTO: NG SOR LUAN

Zhaki Abdullah

Published Aug 24, 2026, 12:30 PM

AgentSea

Synapxe + AWS agentic AI platform for 80,000+ public healthcare professionals — launched end-May 2026.

12,000+ agents

Created by staff in natural language, little or no coding; 300+ shared for reuse.

Real impact

NHCS cardiologist halved pre-consultation prep; KKH agent generates daily rosters.

Guardrails

Patient-data protections built in — "not replaced the need for verification or my own clinical assessment".

- "We are moving from AI as assistant to AI as actor." — SMS Tan Kiat How, HIMSS26 APAC, 24 Aug 2026

Source: The Straits Times, "New platform to create AI agents available to all public healthcare professionals", Zhaki Abdullah, 24 Aug 2026.

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Singapore's Agentic AI Playbook

IMDA framework

Model AI Governance Framework for Agentic AI — launched 22 Jan 2026 (Davos); updated v1.5 May 2026.

Four dimensions

Bound risks upfront · meaningful human accountability · technical controls · end-user responsibility.

Public service

GovTech AI Assistant Desk with an agent registry (owner and purpose of each agent) for ~150,000 officers.

Healthcare

AgentSea plus MOH's AI in Healthcare Guidelines (AIHGle 2.0) — healthcare is a national AI mission.

TIP Course link: approval gates, budgets and audit logs (Labs 3, 7–10) are the 'human accountability' dimension in practice.

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AI for Science — Agents Propose, Humans Decide

Use case	What the agent does	What the human does	Result
Google AI co-scientist (2025)	Multi-agent system generates, debates and ranks hypotheses	Sets the goal, picks hypotheses, runs wet-lab tests	3 of 5 lab-tested leukaemia drug ideas worked
Edison Kosmos (2025)	Reads ~1,500 papers, writes ~42,000 lines of analysis code per run	Sets objective, judges which findings matter	79% of statements judged accurate
Sakana AI Scientist (2025–26)	Idea → code → experiments → paper → review	Chooses the field, checks work, runs ethics review	1 of 3 papers passed a workshop review

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AI for Mathematics

Use case	What the agent does	What the human does	Result
AlphaProof + AlphaGeometry (2024)	Searches formal proofs in Lean	Formalises problems; graders score	IMO silver standard, 28/42
Gemini Deep Think (2025)	Writes natural-language proofs in 4.5 h	IMO coordinators grade and certify	IMO gold, 35/42 (certified)
AlphaEvolve (2025)	Evolves programs scored by an evaluator	Defines the problem and WRITES the evaluator	Improved best-known results in ~20% of 50+ problems
Terence Tao + AI (2025–26)	Searches constructions; proposes proofs (Erdős problems)	Chooses problems, checks and generalises	Real wins — mostly 'cheap wins', says Tao

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Drug Discovery & Biology

Use case	What the AI does	What the human does	Status
AlphaFold 3 (2024)	Predicts structures of proteins, DNA, RNA and ligands	Forms hypotheses, chooses targets, validates in the lab	2024 Nobel Prize in Chemistry (AlphaFold)
Insilico rentosertib (2025)	Found the TNIK target and designed the molecule	Clinicians ran the Phase 2a trial; regulators approved	Nature Medicine, Jun 2025 — safety met
Isomorphic Labs (2026)	In-house AI drug-design engine	Chemists, pharma partners, clinical teams	US$2.1B raised; first trials targeted end-2026

TIP Agents shorten discovery; humans still design the experiment, validate, and carry the clinical and ethical responsibility.

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Business Functions

Function	Agent does	Human does	Evidence
Healthcare	MAI-DxO: virtual panel orders tests step by step	Clinician owns diagnosis and treatment	85.5% on 304 NEJM cases (benchmark only)
Finance	Claude agents for trade accounting and client onboarding	Staff review exceptions and approve	Goldman Sachs + Anthropic, Feb 2026
Software	Codex writes code, tests, CI and docs	Engineers steer, review and build feedback loops	~1M lines, no hand-written code (OpenAI)
Customer service	Handles routine chats in 35+ languages	Takes complex and sensitive cases	Klarna re-prioritised human access in 2025
Marketing	Drafts, designs, schedules and reports campaigns	Sets strategy, approves everything public	Labs 2–5 and 8–9 in this course

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The Pattern Behind Every Success

The agent

- Searches, generates, runs and checks — at scale
- Works 24/7 and in parallel
- Follows the rules it is given — exactly
- Never owns the consequence

The human

- Sets the goal and defines 'good' (rubric, evaluator, tests)
- Approves consequential actions
- Interprets meaning and context
- Owns accountability — always

TIP Counter-evidence matters: a 2025 METR trial found experienced developers 19% SLOWER with early AI tools. The harness and the human make the difference.

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Five Challenges for Trustworthy Agents

Toward agents that are trustworthy, controllable and sustainable.



Hallucination

Confident but false facts → require citations and verifiers.



Security risks

Prompt injection, poisoned skills, leaked keys → least privilege, secrets, sandboxes.



Explainability

Why did it do that? → logs, Kanban history, audit trail.



Data & bias

Fairness, diversity, privacy (PDPA) → mask personal data, review outputs.



Efficiency & cost

Tokens and runs add up → budgets, cheap models for simple roles.

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Five Principles of AI Agent Security

Data control

WHAT AI CAN SEE — classify data; give the agent only what it needs.

Permission control

WHAT AI CAN DO — least privilege; read before write; approve before delete.

Human approval

WHEN AI MUST ASK — gates for public, financial or irreversible actions.

Monitoring & audit

WHAT AI DID — logs, costs, activity trail; review regularly.

Kill switch

HOW TO STOP IT — pause agents, revoke keys, `hermes pause`, Paperclip pause.

TIP Every lab applies at least one of these — look for them.

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The Two Platforms You Will Use

"If OpenClaw is an employee, Paperclip is the company." Hermes is the employee that learns.



Hermes Agent

Day 1 · Labs 1–5

- Open source (MIT), Nous Research
- Self-improving: memory + skills it writes itself
- Telegram & 15+ channels, cron scheduler
- Profiles + Kanban for multi-agent teams

Paperclip

Day 2 · Labs 6–10

- Open source (MIT), self-hosted
- Company: Board → CEO → specialists (org chart)
- Issues, routines, heartbeats
- Budgets, approval gates, immutable audit log

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Activity — Human–AI Workforce Canvas

In pairs, 15 minutes. Then start the Hermes Agent install so it is ready for Lab 1.

1. List the work

Write 8–10 tasks in Kopi Kaki's #ShiokColdBrew campaign (research, posts, video, replies, reporting …).

2. Single agent?

Mark tasks one Hermes agent can do end to end.

3. Agent team?

Mark tasks that need several specialised agents (pipeline or swarm).

4. Human?

Mark decisions that must stay human — and why (risk, accountability, judgement).

5. Checkpoints

Draw where the human approves: before public posts, refunds, spend.

6. Start Hermes

Run the installer now (Lab 1 step 1) — it downloads while we debrief.

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Optional — Ask Hermes to Critique Your Canvas

PROMPT · copy from the Lab Prompts page

```
Here is our Human-AI Workforce Canvas for Kopi Kaki Co.'s
#ShiokColdBrew campaign:
<paste your task list with SINGLE / TEAM / HUMAN labels and
approval checkpoints>

Act as an operations consultant. For each task:
1. Agree or disagree with our label and say why (risk,
reversibility, volume, judgement).
2. Suggest the autonomy level (L1 Operator ... L5 Observer).
3. Name one control (approval gate, budget, audit, verifier) we
must add.
End with the 3 biggest risks in our design.
```

Uses Hermes

Once installed, paste this into the Hermes chat — your first agent conversation.

Compare

Where does the agent disagree with you? Who is right?

Keep it

You will build this exact workforce in Labs 1–10.

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TOPIC 2

Hermes Agent — From a Single AI Employee to a Multi-Agent Team

Day 1 — from one AI employee to a supervised multi-agent team.

1. Lab 1 · Your first AI employee
2. Lab 2 · End-to-end social media campaign
3. Lab 3 · Human-in-the-loop publishing
4. Lab 4 · Multi-agent video studio
5. Lab 5 · Campaign command centre

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What Is Hermes Agent?

"The agent that grows with you." — 255k+ GitHub stars, v0.22 (Oct 2026).

Open source, self-hosted

MIT licence by Nous Research. Runs on your laptop, a VPS, Docker or the desktop app.

Grows with you

Persistent memory and session search; writes and improves its own skills after hard tasks.

Every channel

One gateway: CLI, Telegram, WhatsApp, Slack, Discord, Signal, email …

Tools & MCP

Web search, files, terminal, browser, image generation, TTS and any MCP server.

Proactive

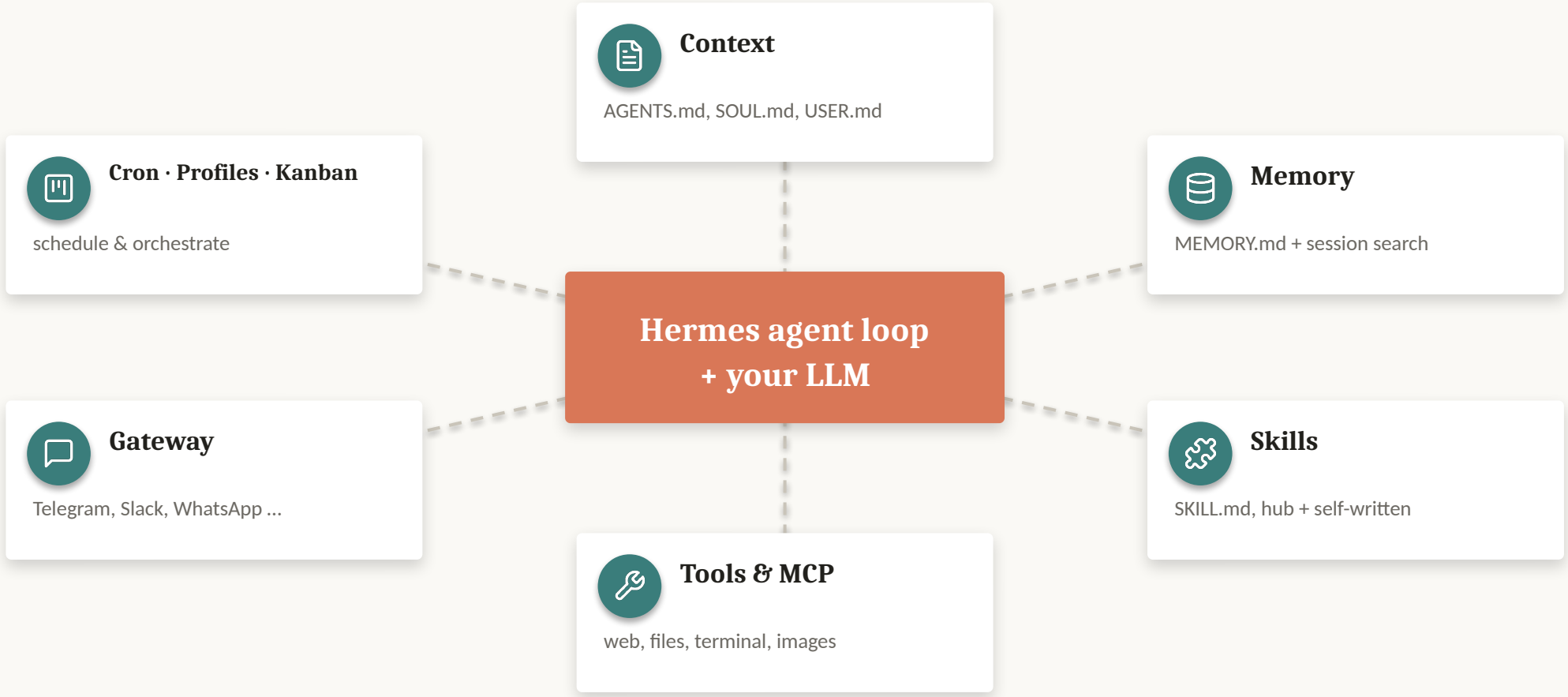
Built-in cron scheduler runs jobs and delivers results to your chat.

Multi-agent

Profiles (isolated agents) + a Kanban board with dependencies, reviews and swarms.

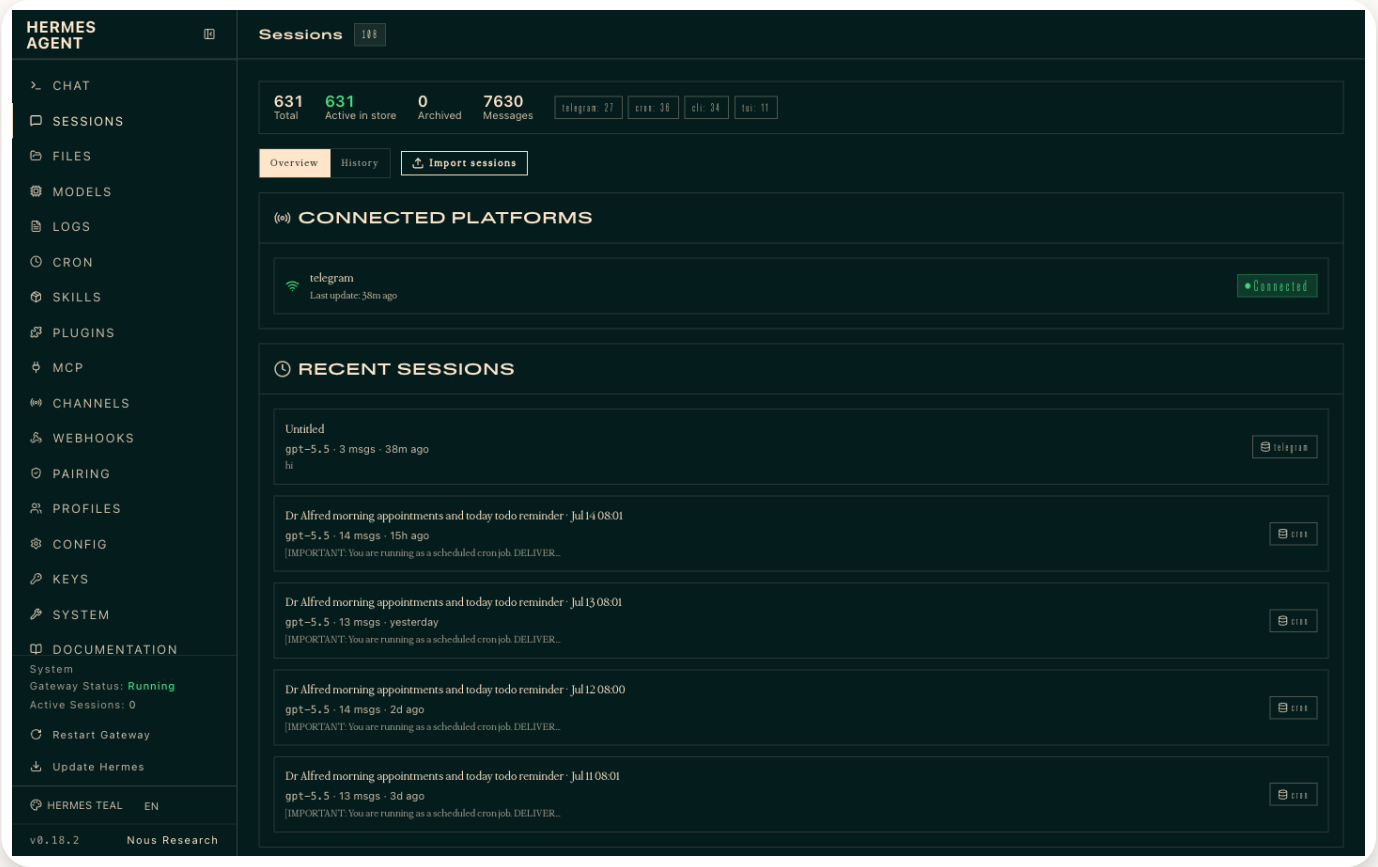
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Hermes Architecture — the Harness Around the Model



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The Hermes Dashboard



- `hermes dashboard` opens http://127.0.0.1:9119 — sessions, skills, profiles, Kanban, cron, channels and keys in one place.

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TOPIC 2 · DAY 1 JOURNEY

Day 1: Single Agent → Multi-Agent → Human Gates

STEP 1

Lab 1

First AI employee — research brief, memory.

STEP 2

Lab 2

Social media campaign — skills + tools.

STEP 3

Lab 3

Human-in-the-loop on Telegram + cron.

STEP 4

Lab 4

5-agent video studio with a human gate.

STEP 5

Lab 5

Swarm analysis, dashboard, decisions.

TIP Each lab saves to ~/kopi-kaki/outputs/labNN — the next lab builds on it.

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The Kopi Kaki Company Pack (data/ in every lab)

File	What it is	Used in
AGENTS.md	Company context Hermes loads automatically from the folder	All labs
brand-guide.md · campaign-brief.md	Voice, colours, formats · product, offer, KPIs, timeline	All labs
compliance-rules.md · video-qa-checklist.md	C1–C10 brand-safety rules · V1–V9 video checks	Labs 2–5, 8–9
customer-reviews.csv · competitors.csv	60 synthetic reviews · 6 fictional competitors	Labs 1, 5
social-comments.csv	20 comments incl. complaints, legal, PII, spam	Labs 3, 5
campaign-week1-metrics.csv · sales-by-outlet.csv	Post metrics and outlet sales for week 1	Lab 5
skills/ · souls/ · tasks/ · DESIGN.md	Ready-made skills, agent roles, task briefs, visual identity	Labs 2, 4, 5

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TOPIC 2 · LAB 1

Lab 1 — Hire Your First AI Employee — Customer & Market Research Brief

Install Hermes, give it company context and long-term memory, and let one agent turn raw reviews and live web research into a brief you can act on.

The business scenario

You are Kopi Kaki Co.'s Marketing Lead. The Gula Melaka Cold Brew launches on 1 Dec and you have 60 unread customer reviews, a competitor list and no analyst. You 'hire' Mia — a Hermes Agent marketing analyst — onboard her with the company context, and ask for a research brief that tells you which campaign angle to run.

You will deliver

✓ outputs/lab01/review-insights.md — ratings, themes and evidence (review IDs)
✓ outputs/lab01/research-brief.md — trends, competitors, 3 campaign angles with evidence
✓ Mia's long-term memory holds the brand facts and YOUR chosen angle (recalled in a new session)

Agents 1 Hermes agent — 'Mia', AI marketing analyst

Human role Set the goal, challenge the evidence, choose the campaign angle

Time 55 min · Single agent · labs/lab01-hermes-first-ai-employee/

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Lab 1 — How the workflow runs

Perception → reasoning → action, with you at the checkpoints.

AGENT · STEP 1

Install & onboard

Install Hermes, pick a ≥64k-context model, load AGENTS.md.

AGENT · STEP 2

Remember the brand

Save brand facts to long-term memory.

AGENT · STEP 3

Perceive

Read 60 reviews + search the web for 2026 trends.

AGENT · STEP 4

Reason

Find themes, compare competitors, draft 3 angles.

HUMAN · STEP 5

Decide

Challenge one claim, choose the angle.

AGENT · STEP 6

Recall

New session: Mia remembers your decision.

Checkpoint Every insight cites review IDs or a URL; the angle you chose is recalled in a brand-new session.

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Install, Connect a Model, Start in the Workspace

TERMINAL · copy from the Lab Prompts page

```
curl -fsSL https://hermes-agent.nousresearch.com/install.sh |
bash
# Windows PowerShell:  iex (irm https://hermes-
agent.nousresearch.com/install.ps1)
source ~/.zshrc    # or ~/.bashrc — puts 'hermes' on your PATH
```

Install

macOS / Linux / WSL2 one-liner; Windows PowerShell installer.

hermes setup

Pick a provider and a model with ≥ 64k context; `hermes doctor` to check.

Start in ~/kopi-kaki

AGENTS.md there is loaded automatically.

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Context & Memory Files in Hermes

File	Scope	What goes in it	Loaded
AGENTS.md (project)	This folder	Company facts, rules, current priority	Every session started in the folder
SOUL.md	~/.hermes/ (per profile)	The agent's role, personality and standing rules	Every session
MEMORY.md	~/.hermes/memories/	Facts the agent decides to remember (~2,200 chars)	Session start
USER.md	~/.hermes/memories/	What it knows about you and your preferences	Session start
Session history	state database	Every past conversation — searchable	On demand (session search)

TIP Short-term = this session's context window. Long-term = the files and history above.

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TOPIC 2 · LAB 1

Onboard Mia — persona and long-term memory

PROMPT · copy from the Lab Prompts page

```
You are Mia, Kopi Kaki Co.'s AI marketing analyst. Read
AGENTS.md, brand-guide.md and campaign-brief.md in this folder.

Save these facts to your long-term memory:
- Kopi Kaki Co.: Singapore kopi chain, 12 outlets, positioning
"Heritage kopi, modern craft".
- Launch: Gula Melaka Cold Brew, SGD 6.80, launches 1 Dec 2026,
campaign #ShiokColdBrew 17 Nov - 14 Dec.
- Promo SHIOK20: 20% off the first Gula Melaka Cold Brew ordered
in the app, valid 1-14 Dec 2026, one per customer.
- Never publish anything without my APPROVE. Never make health or
halal claims.

Then summarise the company in 5 bullets and ask me the 3
questions a new analyst should ask on day one.
```

Paste into

Hermes chat (started inside ~/kopi-kaki)

Expect

Mia confirms the memory was saved and returns 5 bullets + 3 questions. Check the file itself: `cat ~/.hermes/memories/MEMORY.md` shows the facts.

You decide

Set the goal, challenge the evidence, choose the campaign angle

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Perceive the data — analyse 60 customer reviews

PROMPT · copy from the Lab Prompts page

```
Analyse customer-reviews.csv.
1. Rating distribution (count and % for 1-5 stars) and average
rating per outlet.
2. The top 5 praise themes and top 5 complaint themes, each with
a count and 2 example review_ids.
3. Anything relevant to a cold-brew launch (mentions of cold
brew, sweetness, oat milk, app).
Show your calculation method. Save the result as
outputs/lab01/review-insights.md.
```

Paste into

Hermes chat

Expect

A Markdown report with a ratings table and themes that cite review IDs (e.g. R1012).

You decide

Set the goal, challenge the evidence, choose the campaign angle

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Lab 1 — Checkpoint & Reflection

Verify

outputs/lab01/review-insights.md and research-brief.md exist and cite review IDs / URLs

Verify

The brief ends with a decision log naming the angle you chose

Verify

MEMORY.md contains the brand facts and your decision

Concepts in action

Perception (files + web) → Brain (LLM reasoning, memory) → Action (write files) · AGENTS.md = context the agent loads automatically (context engineering)

Stretch challenge

Ask Mia: "Save the review-analysis procedure you just used as a reusable skill called review-insights." Then run `hermes skills list` — a self-improving agent turns experience into procedural memory.

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TOPIC 2 · LAB 2

Lab 2 — End-to-End Social Media Campaign with One Agent

Give the agent two company skills and one tool chain, and turn the brief into a complete, dated, on-brand multi-platform campaign — then improve it with your feedback.

The business scenario

With the angle chosen, you need a campaign: strategy, a 14-day content calendar across Instagram, TikTok, Facebook and LinkedIn, captions in the Kopi Kaki voice, visuals and tracking links. Instead of a long prompt you give Mia two skills — the company's brand-voice skill and a campaign-planner skill — so the work is repeatable and self-checked.

You will deliver

✓ outputs/lab02/campaign-plan.md — strategy, pillars, phases, KPIs, budget split
✓ outputs/lab02/content-calendar.csv — ≥ 16 dated posts with UTM links
✓ outputs/lab02/captions.md — captions with a compliance self-check per post
✓ outputs/lab02/images/ — 3 hero visuals (or image prompts if no image tool)
✓ A revised v2 after your feedback

Agents 1 Hermes agent (Mia) + 2 custom skills + image tool

Human role Review the draft, give concrete feedback, accept v2

Time 60 min · Single agent + skills · labs/lab02-hermes-social-media-campaign/

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Lab 2 — How the workflow runs

Perception → reasoning → action, with you at the checkpoints.

AGENT · STEP 1

Install skills

Add kopi-brand-voice and social-campaign-planner.

AGENT · STEP 2

Plan

Strategy, pillars, phases, KPI mapping.

AGENT · STEP 3

Calendar

16+ dated posts, platform formats, UTMs.

AGENT · STEP 4

Create

Captions + hero visuals with self-check.

HUMAN · STEP 5

Review

You give 3 concrete change requests.

AGENT · STEP 6

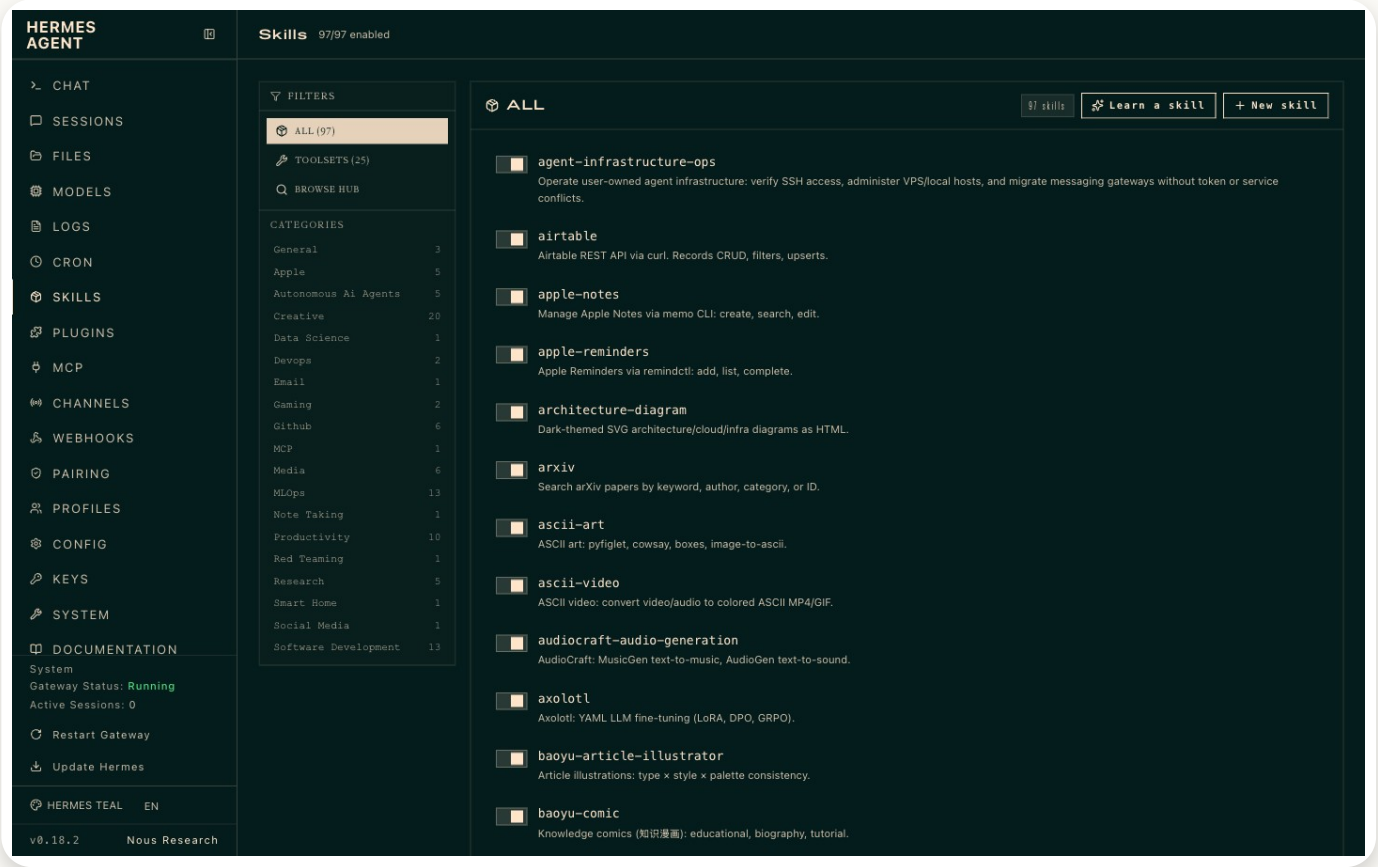
Revise

v2 applies the feedback and re-checks.

Checkpoint Every caption carries a PASS self-check; no health/halal claims; SHIOK20 always shows its terms.

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Skills on the Hermes Dashboard



- Skills are folders with a SKILL.md — bundled, from the hub (`hermes skills install`), your own, or written by the agent itself.

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Anatomy of a Skill — kopi-brand-voice/SKILL.md

SKILL.md · in labs/lab02-.../data/skills/

```
---
name: kopi-brand-voice
description: Write or rewrite public-facing copy in the
  Kopi Kaki voice and check it against the compliance rules.
---
# Kopi Kaki brand voice
## Steps
1. Read brand-guide, campaign-brief, compliance-rules.
2. Warm and sensory; max 1 Singlish word, max 2 emojis.
3. Only facts and prices from the brief or catalogue.
4. Promo code shown? Add its terms.
5. Run the self-check; fix every FAIL.
## Self-check: C1 health · C3 promo · C4 halal · C9 AI
```

description

Tells the agent WHEN to load the skill — costs no context until used.

steps

An SOP the agent follows the same way every time.

self-check

Verification built into the skill — the loop checks itself.

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Build the campaign plan and calendar

PROMPT · copy from the Lab Prompts page

```
Use the social-campaign-planner skill to build the #ShiokColdBrew
campaign from campaign-brief.md, using the angle I chose in Lab 1
(check your memory).
Scope: teaser (17-23 Nov) and pre-order push (24-30 Nov) phases,
Instagram, TikTok, Facebook and LinkedIn.
Deliver:
- outputs/lab02/campaign-plan.md
- outputs/lab02/content-calendar.csv with at least 16 posts (use
the columns in the skill)
- outputs/lab02/captions.md with every caption checked by the
kopi-brand-voice skill
End with a 5-line summary of decisions you need from me.
```

Paste into

Hermes chat in ~/kopi-kaki

Expect

Three files in outputs/lab02 and a decisions summary.

You decide

Review the draft, give concrete feedback, accept v2

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Create the hero visuals

PROMPT · copy from the Lab Prompts page

```
Create 3 hero visuals for the teaser posts using your image
generation tool: square 1:1, brand colours (kopi brown #5B3A29,
pandan green #4E8A3E, cream #F6EBD9, gula melaka amber #C7782A),
natural light, a glass of iced cold brew with a gula melaka
drizzle, kopitiam tiles, no people's faces, no text in the image.
Save them to outputs/lab02/images/ and add the file names to
content-calendar.csv. If you have no image tool, write the 3
prompts to outputs/lab02/images/prompts.md instead.
```

Paste into

Hermes chat

Expect

3 PNGs (or a prompts file) linked from the calendar.

You decide

Review the draft, give concrete feedback, accept v2

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Lab 2 — Checkpoint & Reflection

Verify

campaign-plan.md, content-calendar(-v2).csv and captions(-v2).md exist in outputs/lab02

Verify

≥ 16 posts across 4 platforms with UTM links; LinkedIn has no promo code

Verify

Each caption has a PASS self-check table

Concepts in action

Skills = procedural memory: a SKILL.md the agent loads on demand · Tools = actions: file write, web search, image generation

Stretch challenge

Ask Mia to turn your three review comments into permanent rules by patching the kopi-brand-voice skill ("update the skill so this mistake never happens again") — that is harness engineering in one sentence.

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TOPIC 2 · LAB 3

Lab 3 — Human-in-the-Loop Publishing & Community Care (Telegram + Cron)

Put the agent on Telegram, make it ask for your APPROVE before anything goes public, schedule it every morning, and let it triage comments while escalating the risky ones to you.

The business scenario

The campaign is planned; now it must run every day without you babysitting it. Each morning Mia sends you the day's post and image on Telegram. Only when you reply APPROVE does she publish it to the brand's Telegram channel and log it. She also works through the comment inbox — answering simple questions, but escalating food-safety, allergy, legal, billing, media and halal questions to you with personal data masked.

You will deliver

- ✓ A Telegram bot connected to Hermes (your private chat with Mia)
- ✓ A 'Kopi Kaki Feed (Training)' Telegram channel where approved posts are published
- ✓ outputs/lab03/approval-log.csv — every APPROVE / EDIT / REJECT decision
- ✓ A daily 09:00 cron job that sends the day's post for approval
- ✓ outputs/lab03/moderation-queue.csv + an escalation summary on Telegram

Agents 1 Hermes agent (Mia) on the messaging gateway + cron scheduler

Human role Approve, edit or reject every post; decide every escalation

Time 45 min · Human–agent collaboration · labs/lab03-hermes-human-in-the-loop/

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Lab 3 — How the workflow runs

Perception → reasoning → action, with you at the checkpoints.

AGENT · STEP 1

Connect

Telegram bot + allow-listed user ID.

AGENT · STEP 2

Propose

09:00 cron sends the day's post + image.

HUMAN · STEP 3

Approve

You reply APPROVE / EDIT / REJECT.

AGENT · STEP 4

Publish

Approved post goes to the channel; logged.

AGENT · STEP 5

Triage

Answer FAQs, mask PII, escalate risks.

HUMAN · STEP 6

Decide

You resolve each escalation.

Checkpoint Nothing reaches the channel without an APPROVE in approval-log.csv; all 7 risky comments are escalated, none answered.

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Connect Telegram in Five Steps

STEP 1



@BotFather

/newbot → name → copy the token (keep it secret).

STEP 2



@userinfobot

Copy your numeric Telegram user ID.

STEP 3



hermes gateway setup

Choose Telegram, paste token + your ID as the only allowed user.

STEP 4



hermes gateway run

Leave it running — it hosts chat, cron and Kanban dispatch.

STEP 5



Brand channel

Create 'Kopi Kaki Feed (Training)'; add the bot as admin.

TIP Shortcut: Dashboard → Messaging → Telegram → 'Create with QR'. NEVER share the bot token.

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Escalation Rules — Policy Becomes Agent Behaviour

Comment type	Agent action	Why
FAQ (availability, oat milk, outlets)	REPLY — draft in brand voice	Low risk, high volume
Spam	IGNORE	No value; don't engage
Food safety / allergy (C06, C16)	ESCALATE — no public reply	Health and legal risk
Legal threat / media (C18, C11)	ESCALATE	Needs a human spokesperson
Billing / refund (C09)	ESCALATE	Money and policy decisions
Halal / health questions (C03, C15)	ESCALATE	Claims we must not make (C1, C4)

TIP Personal data (names, phone numbers) is masked before anything reaches you or a public reply.

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Make It Proactive — the 09:00 Approval Cron

TERMINAL · copy from the Lab Prompts page

```
hermes cron create "0 9 * * *" "Proposal only - never publish.
Take the next post with status draft from
~/kopi-kaki/outputs/lab02/content-calendar-v2.csv, set it to
pending-approval and send me its post id, image, platform, date
and caption, ending with: Reply APPROVE <post_id>, EDIT
<post_id>: <changes> or REJECT <post_id>." --name daily-post-
approval --deliver telegram
hermes cron list
hermes cron run <job_id>
```

Schedule

Cron syntax '0 9 * * *' = every day at 09:00.

--deliver telegram

Results arrive in your Telegram chat.

Still gated

The job proposes; only your APPROVE publishes.

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Run one approval cycle by hand

PROMPT · copy from the Lab Prompts page

```
Approval workflow for #ShiokColdBrew (standing rule - add it to
~/kopi-kaki/AGENTS.md too):
1. Take the next post with status "draft" from
~/kopi-kaki/outputs/lab02/content-calendar-v2.csv, set its status
to "pending-approval" and send it to me here with its post id,
image, platform and date. End with: Reply APPROVE <post_id>, EDIT
<post_id>: <changes> or REJECT <post_id>.
2. Whenever I reply APPROVE <post_id>, read that row from the CSV
- it must say "pending-approval" - then post the caption and
image to telegram:<channel_id>, set the status to "published" and
append a row to ~/kopi-kaki/outputs/lab03/approval-log.csv.
3. EDIT: apply my changes, re-check with kopi-brand-voice, keep
"pending-approval" and send it again. REJECT: set "rejected" and
log it.
4. Never publish a row that is not "pending-approval" and never
without my APPROVE.
```

Paste into

Telegram → your bot

Expect

You receive the draft with its post id; after APPROVE <post_id> it appears in the channel, the CSV says published and approval-log.csv has a row.

You decide

Approve, edit or reject every post; decide every escalation

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TOPIC 2 · LAB 3

Triage the comment inbox with escalation rules

PROMPT · copy from the Lab Prompts page

```
Triage ~/kopi-kaki/social-comments.csv using compliance-rules.md.
For each comment decide: REPLY (safe FAQ), IGNORE (spam) or
ESCALATE (food safety, allergy, legal threat, billing/refund,
media enquiry, halal question, health claim question, anything
uncertain).
- For REPLY: draft a reply in the kopi-brand-voice skill.
- For ESCALATE: do NOT draft a public reply. Mask all personal
data (names, phone numbers) and recommend the next step.
Save outputs/lab03/moderation-queue.csv (comment_id, decision,
reason, draft_reply) and send me an escalation summary here.
```

Paste into

Telegram → your bot (or Hermes chat in ~/kopi-kaki)

Expect

7 escalations summarised on Telegram with phone numbers masked; safe replies drafted.

You decide

Approve, edit or reject every post; decide every escalation

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Lab 3 — Checkpoint & Reflection

Verify

Bot replies only to your allow-listed user ID

Verify

Every published post has an APPROVE row in approval-log.csv and its CSV status went pending-approval → published

Verify

`hermes cron list` shows daily-post-approval

Concepts in action

Human-in-the-loop (approve before acting) vs human-on-the-loop (monitor and intervene) · Escalation rules turn policy into agent behaviour

Stretch challenge

Add a second gate: a post that mentions a price or promo must be approved twice (you and a colleague's Telegram ID). How would you express that rule, and what does it cost in speed?

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TOPIC 2 · LAB 4

Lab 4 — Multi-Agent Video Studio — Research → Script → Production → Audit → Publish

Build a five-agent production team on the Kanban board, render a real 25-second vertical video with HyperFrames, let an auditor agent check it, and approve it yourself before it publishes.

The business scenario

The campaign needs a 25-second TikTok/Reels launch video. One agent doing everything is slow, expensive and hard to review, so you staff a studio: a researcher finds the hook, a scriptwriter writes it, a producer renders it, an auditor checks it against the brand and compliance rules, and a publisher prepares and posts it — but only after you approve the cut.

You will deliver

✓ 5 Hermes profiles with routing descriptions (researcher, scriptwriter, producer, auditor, publisher)
✓ A Kanban task graph with dependencies and a human gate before publishing
✓ video-studio/research.md, script.md (scene list), output/final.mp4 (9:16)
✓ video-studio/audit-report.md — PASS/FAIL on V1–V9 and C1–C10
✓ video-studio/publish-package.md, reviewed by you, then the video posted once to your Telegram channel

Agents 5 Hermes profiles coordinated by the Kanban board

Human role Review the cut and the publish package, then release the posting task (or send it back with comments)

Time 60 min · Multi-agent (ordered pipeline) · labs/lab04-hermes-multi-agent-video-studio/

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Lab 4 — How the workflow runs

Perception → reasoning → action, with you at the checkpoints.

AGENT · STEP 1

Research

Hooks, trends, proof points (cited).

AGENT · STEP 2

Script

25 s script + 6-scene list.

AGENT · STEP 3

Produce

HyperFrames render → final.mp4.

AGENT · STEP 4

Audit

V1–V9, C1–C10 → PASS / FAIL.

HUMAN · STEP 5

Approve

Review cut + package; unblock.

AGENT · STEP 6

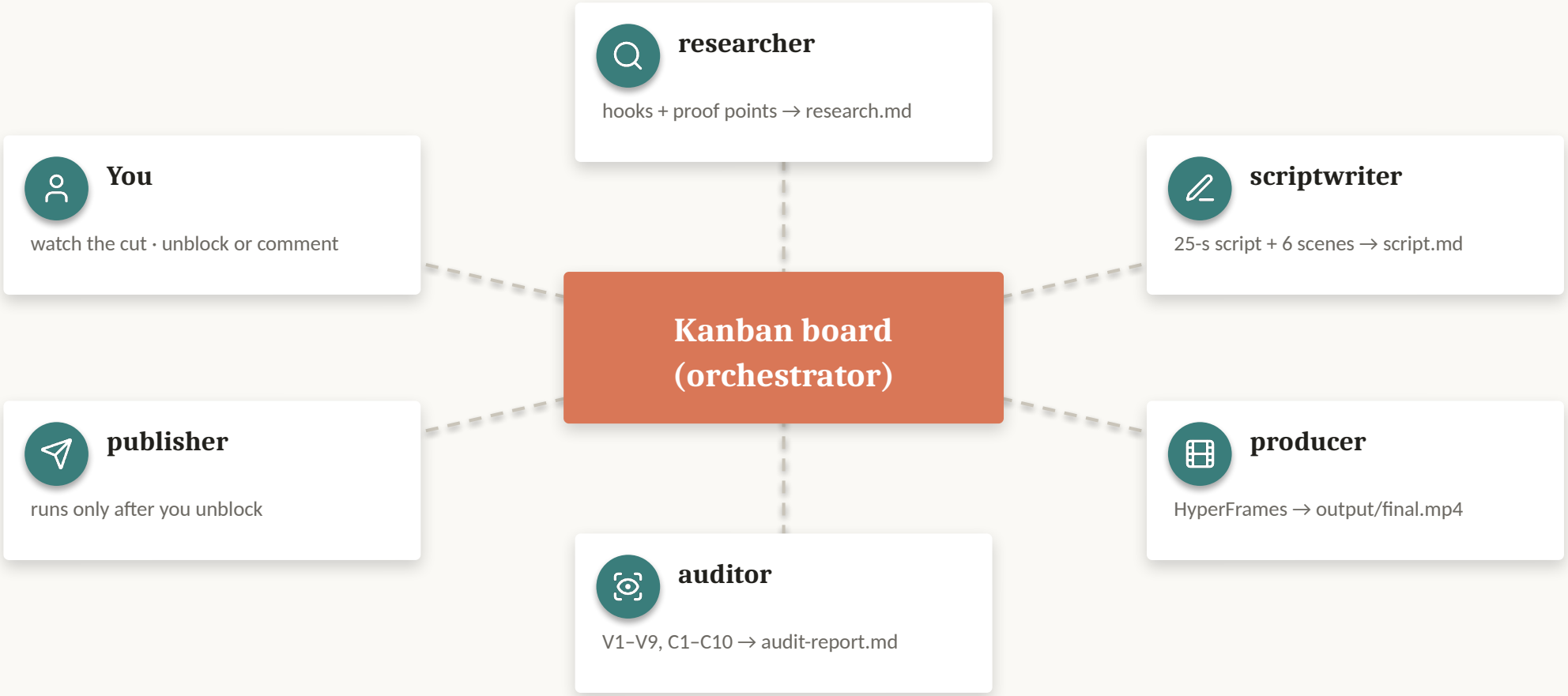
Post

Posted once to the channel.

Checkpoint final.mp4 is 9:16 and 20–30 s, the audit is PASS, and the posting task only ran after you reviewed the package and unblocked it.

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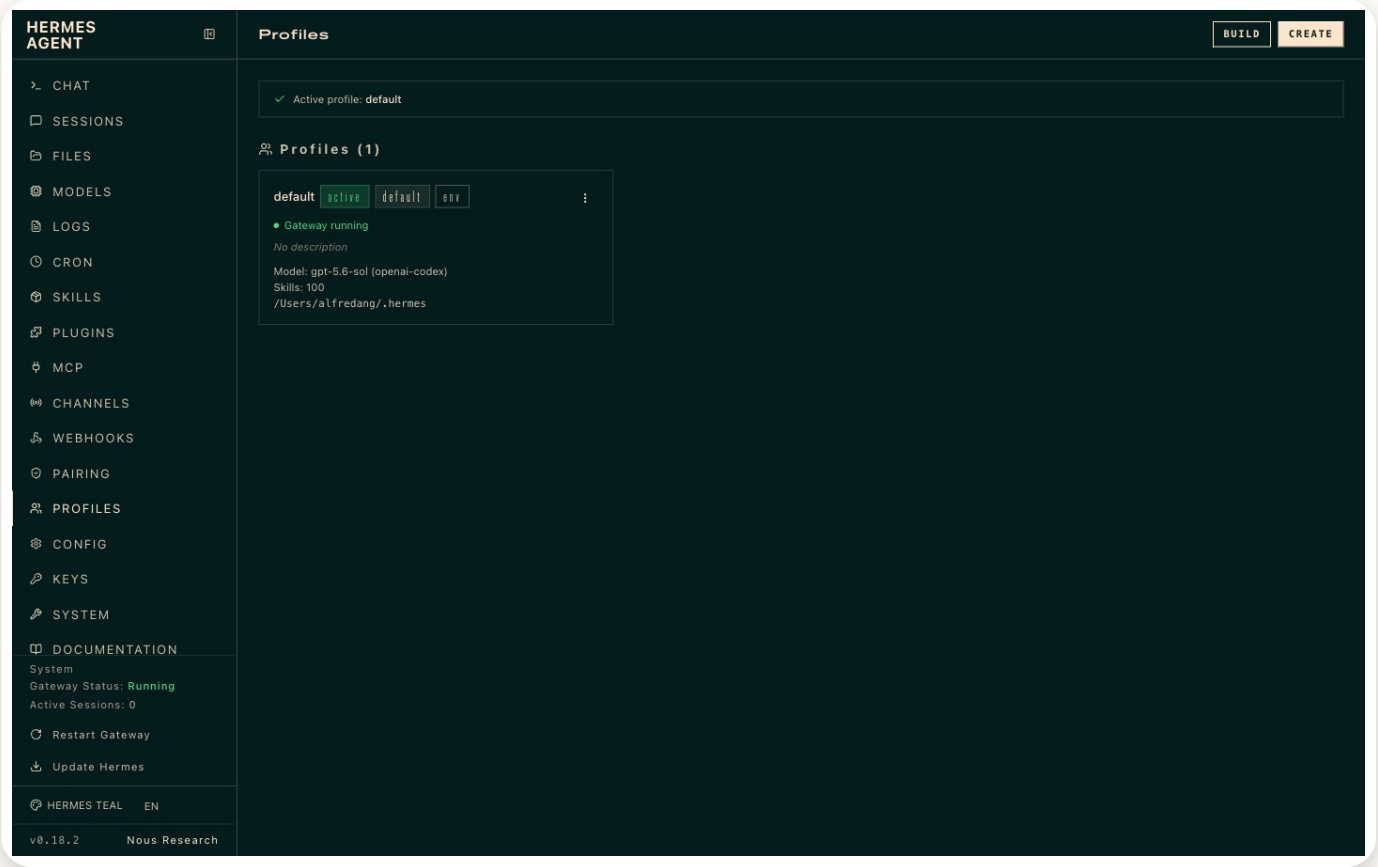
Your Video Studio Team



TIP The profile --description is the routing contract; SOUL.md holds each role's rules.

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Profiles — Isolated Agents with Their Own Memory



- `hermes profile create <name> --clone --description "..."` — each profile has its own SOUL.md, memory, skills and sessions.

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Build the Task Graph with a Human Gate

TERMINAL · copy from the Lab Prompts page

```
S=dir:$HOME/kopi-kaki/video-studio
hermes kanban create "Research hooks for the Gula Melaka Cold Brew launch
short" --assignee researcher --workspace $S --tenant kopi-video --body-
file ~/kopi-kaki/tasks/t1-research.md
hermes kanban create "Write the 25-second script and scene list" --
assignee scriptwriter --parent <T1> --workspace $S --tenant kopi-video --
body-file ~/kopi-kaki/tasks/t2-script.md
hermes kanban create "Render the 9:16 launch video with HyperFrames" --
assignee producer --parent <T2> --workspace $S --tenant kopi-video --
skill hyperframes --max-runtime 45m --body-file ~/kopi-kaki/tasks/t3-
produce.md
hermes kanban create "Audit the video and copy (V1-V9, C1-C10)" --
assignee auditor --parent <T3> --workspace $S --tenant kopi-video --skill
brand-compliance-audit --body-file ~/kopi-kaki/tasks/t4-audit.md
hermes kanban create "Build the publish package (do not post)" --assignee
publisher --parent <T4> --workspace $S --tenant kopi-video --body-file
~/kopi-kaki/tasks/t5-package.md
hermes kanban create "Post the approved video to the channel" --assignee
publisher --parent <T5> --workspace $S --tenant kopi-video --initial-
status blocked --body-file ~/kopi-kaki/tasks/t6-post.md
```

--parent

Each task waits for the one before — an ordered SOP.

dir: workspace

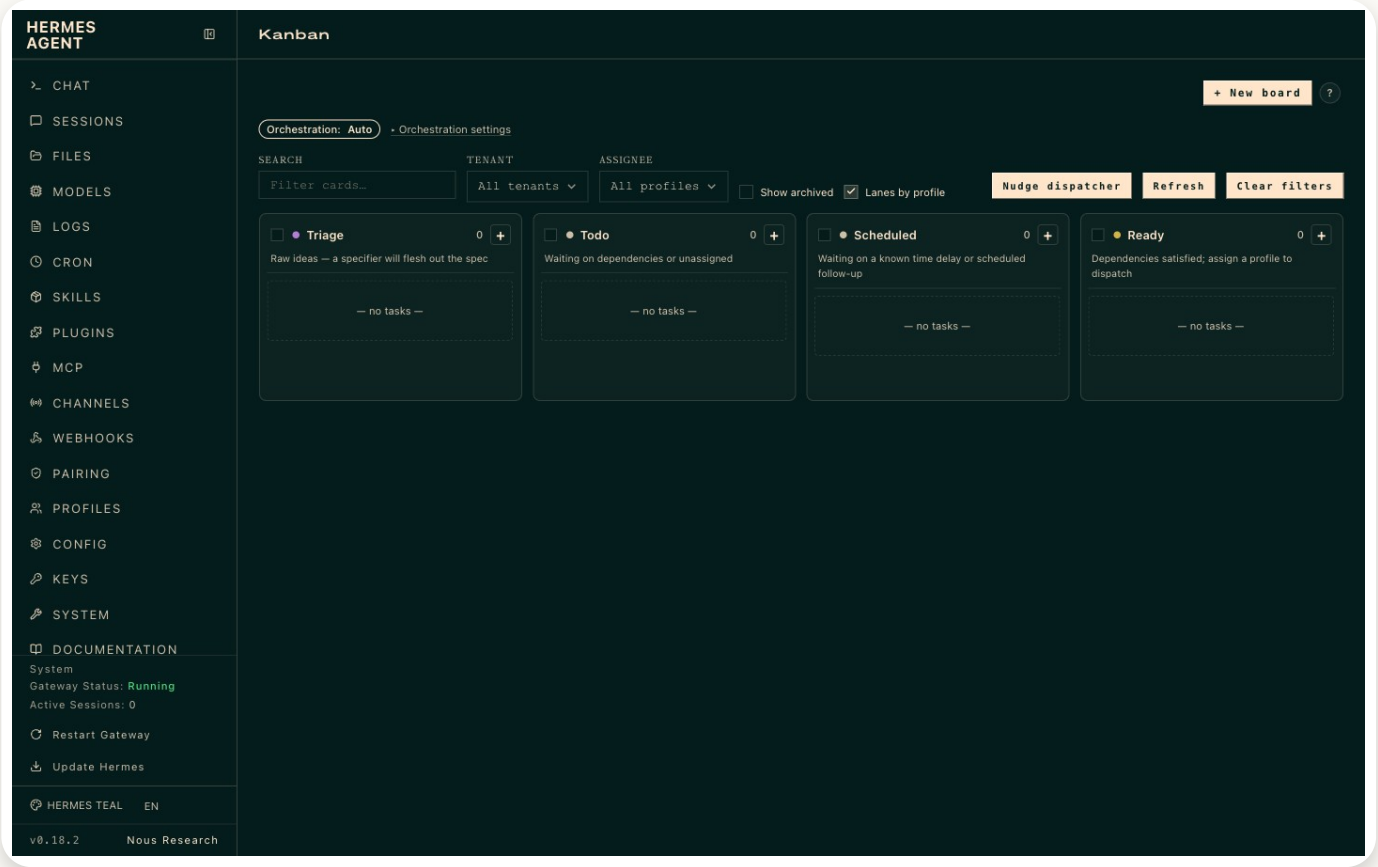
All stages share ~/kopi-kaki/video-studio.

--initial-status blocked

Publishing waits for your `hermes kanban unblock`.

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Watch the Board



- Lanes: triage → todo → ready → running → review → done. A FAIL goes back with `hermes kanban request-changes`.

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Reflect — why a team beats one big agent

PROMPT · copy from the Lab Prompts page

```
Read the Kanban history for tenant kopi-video (hermes kanban show
for each task). Report per stage: which profile ran it, how long
it took, how many retries or change requests it needed, and what
was handed to the next stage. Then compare this five-agent design
with a single agent doing everything: quality, cost, speed,
reviewability.
```

Paste into

Hermes chat

Expect

A stage-by-stage table you can use in the assessment.

You decide

Review the cut and the publish package, then release the posting task (or send it back with comments)

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Lab 4 — Checkpoint & Reflection

Verify

`hermes profile list` shows the five roles

Verify

final.mp4 is 1080×1920, 20–30 s (`ffprobe` or the auditor's report)

Verify

audit-report.md is PASS (or shows a FAIL that was fixed via request-changes)

Concepts in action

Agentic AI = several specialised agents + orchestration (vs one AI agent) · Profile description = routing contract; SOUL.md = role + rules

Stretch challenge

Shortcut: install `official/creative/kanban-video-orchestrator`, give it the brief, and compare the team it designs with yours. Which roles did it add or merge?

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TOPIC 2 · LAB 5

Lab 5 — Multi-Agent Campaign Command Centre — Measure, Optimise & Report

Run three analysts in parallel, have a verifier check their numbers, and get a dashboard plus an optimisation plan you approve — then schedule it every Monday.



The business scenario

Week 1 of #ShiokColdBrew is over. Leadership wants to know: are we on track for 2,500 pre-orders, which posts work, what customers are saying and what competitors are doing? You launch a Kanban swarm — three parallel workers, a verifier that re-checks their numbers, and a synthesiser that builds the dashboard and recommendations. You decide which optimisations to apply.

You will deliver

✓ outputs/lab05/dashboard.html — KPI cards + charts (opens in a browser)
✓ outputs/lab05/optimisation-plan.md — 3 proposals with evidence, impact, risk
✓ outputs/lab05/verification.md — the verifier's re-calculation of the key numbers
✓ Your decision recorded on the swarm task
✓ A Monday 09:00 cron that delivers the weekly report to Telegram

Agents 3 parallel workers + verifier + synthesiser (Kanban swarm)

Human role Spot-check one number yourself; approve which optimisations to run

Time 45 min · Multi-agent (parallel swarm) · labs/lab05-hermes-campaign-command-centre/

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Lab 5 — How the workflow runs

Perception → reasoning → action, with you at the checkpoints.

AGENT · STEP 1



Analyse

KPIs vs targets from metrics + sales CSVs.

AGENT · STEP 2



Listen

Sentiment and risks in comments.

AGENT · STEP 3



Watch

Competitor & trend scan (cited).

AGENT · STEP 4



Verify

Re-compute key numbers; flag gaps.

AGENT · STEP 5



Synthesise

Dashboard + optimisation plan.

HUMAN · STEP 6



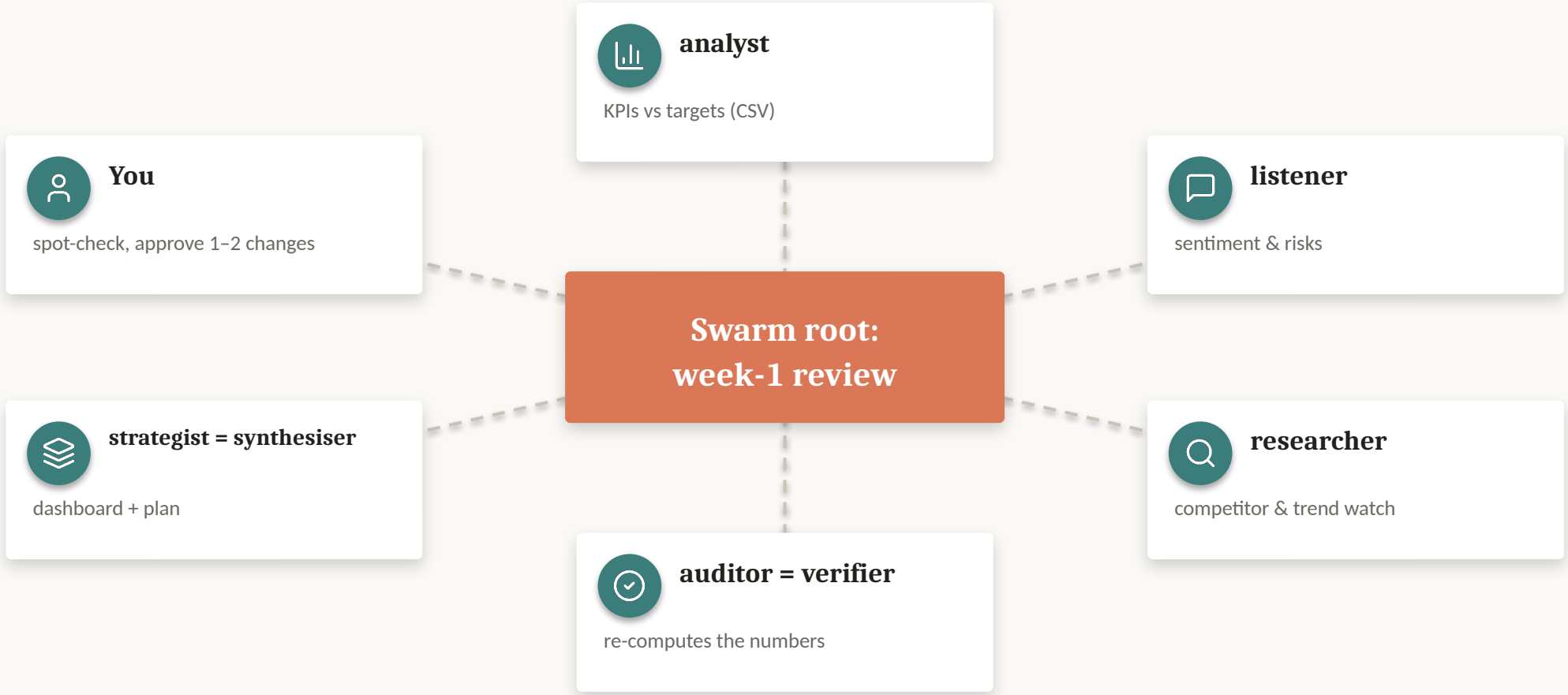
Decide

Spot-check, approve 1–2 changes.

Checkpoint Your own calculation of pre-orders and cost per pre-order matches the dashboard; the plan you approved is recorded on the task.

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Parallel Swarm with a Verifier



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The KPIs You Will Check

KPI	Formula	Target
App pre-orders	Σ app_preorders	2,500 by 30 Nov
Engagement rate	(likes + comments + shares) ÷ reach	≥ 8% (IG + TikTok)
Click-through rate	link_clicks ÷ reach	Trend up week on week
Cost per pre-order	paid_spend_sgd ÷ app_preorders	Lower is better
Video views	Σ video_views (TikTok + Reels)	40,000

TIP Never forward a number you have not checked: verify at least one by hand.

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Launch the Swarm

TERMINAL · copy from the Lab Prompts page

```
hermes kanban swarm "Week-1 #ShiokColdBrew review and
optimisation plan. FIRST read ~/kopi-kaki/tasks/swarm-definition-
of-done.md; data in ~/kopi-kaki; write outputs to
~/kopi-kaki/outputs/lab05/" \
  --worker "analyst:KPIs vs targets from ~/kopi-kaki/campaign-
week1-metrics.csv and sales-by-outlet.csv per the definition of
done" \
  --worker "listener:Sentiment and risk review of
~/kopi-kaki/social-comments.csv per the definition of done" \
  --worker "researcher:Competitor and cold-brew trend watch from
the most recent real sources, dated and cited, per the definition
of done" \
  --verifier auditor --synthesizer strategist
hermes kanban list            # copy the swarm root id
```

--worker

Parallel cards: profile:title.

--verifier

Waits for all workers, re-checks.

--synthesizer

Builds the dashboard and plan.

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Decide and record

PROMPT · copy from the Lab Prompts page

```
I approve proposals <1 and 3> from outputs/lab05/optimisation-
plan.md and reject proposal <2> because <reason>. Comment this
decision on the swarm root card <swarm_root_id>, then update
outputs/lab02/content-calendar-v2.csv for 24-30 Nov to apply the
approved changes (status stays "draft" so each post still goes
through my Telegram approval).
```

Paste into

Hermes chat in ~/kopi-kaki

Expect

Decision comment on the board and an updated calendar.

You decide

Spot-check one number yourself; approve which optimisations to run

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Lab 5 — Checkpoint & Reflection

Verify

dashboard.html opens and shows the four KPIs vs target

Verify

verification.md re-computes at least pre-orders, spend, engagement rate and cost per pre-order

Verify

Your manual check matches (data covers 17–22 Nov)

Concepts in action

Parallel (swarm) vs ordered (pipeline) multi-agent patterns · Verifier agent + human spot-check = two layers of verification

Stretch challenge

Feed the result back into Lab 3: ask Mia to add the approved changes to tomorrow's approval message, so the loop runs plan → publish → measure → improve without leaving Telegram.

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What You Built on Day 1

Single agent

Mia researched, remembered and ran a whole campaign with skills and tools.

Human-in-the-loop

Nothing went public without your APPROVE; risky comments came to you.

Multi-agent

A five-role studio and a verified analysis swarm.

Harness

Context, memory, skills, verification, gates, cron — the environment did the heavy lifting.

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TOPIC 3

Paperclip — Running & Governing a Company of AI Agents

Day 2 — the same marketing work, run as a governed company.

1. Lab 6 · Found the AI company
2. Lab 7 · Hire the team
3. Lab 8 · Launch campaign project
4. Lab 9 · Weekly video routine
5. Lab 10 · Govern, measure & improve

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What Is Paperclip?

"If OpenClaw is an employee, Paperclip is the company." — 100k+ GitHub stars, self-hosted at http://localhost:3100.

The company as the harness

Open-source orchestration for a company of agents — org chart, goals, budgets, governance.

You are the Board

Approve hires and strategy; pause, override, reassign or terminate any agent.

Bring your own agent

Adapters for Claude Code, Codex, Gemini CLI, OpenClaw, Hermes, Cursor …

Issues & goals

Mission → goals → projects → issues: every task traces to a goal.

Heartbeats & routines

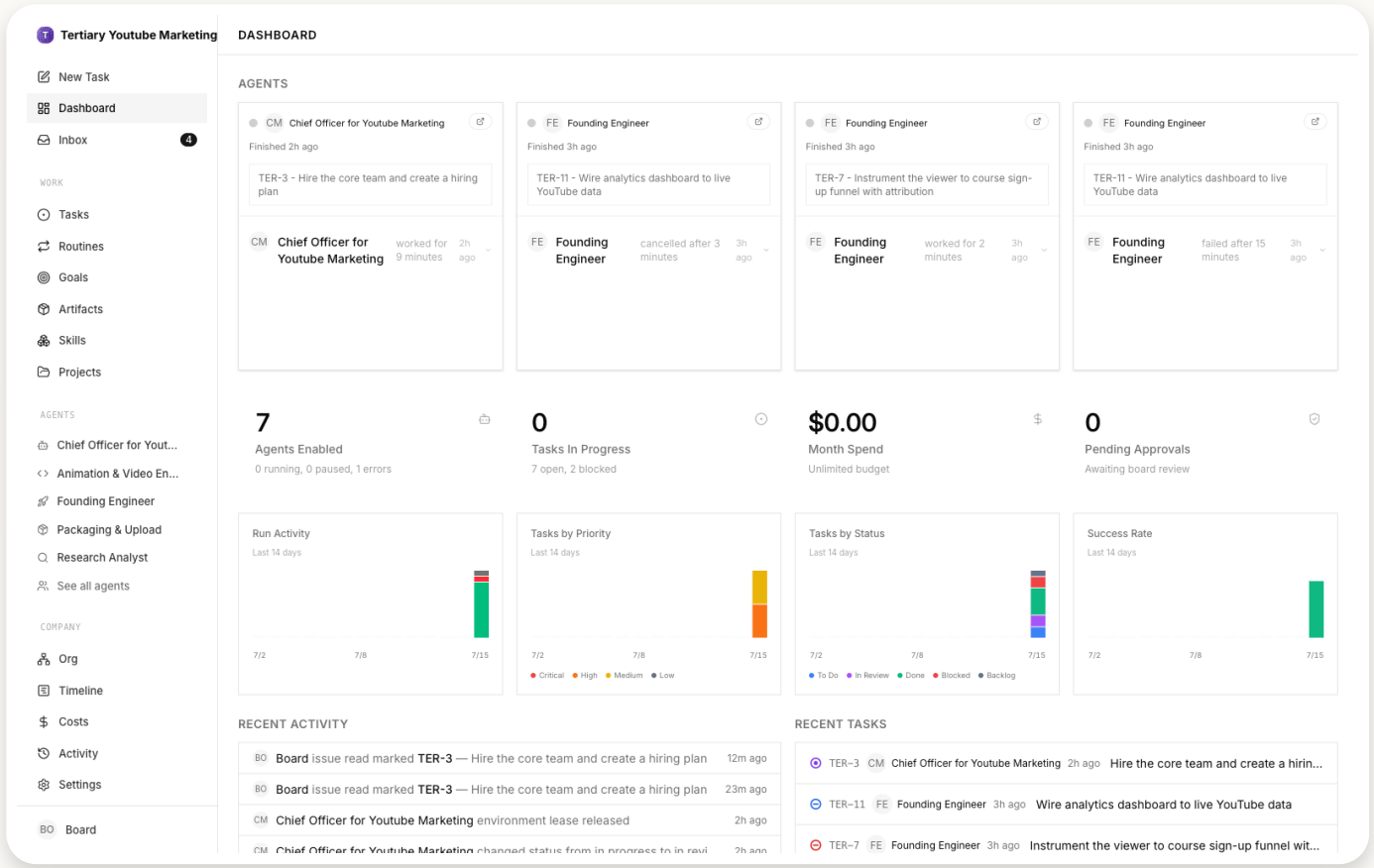
Agents wake on a schedule or when work is assigned; routines create recurring work.

Budgets & audit

Per-agent budgets warn at 80% and auto-pause at 100%; append-only activity log.

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The Paperclip Dashboard



- Agents, issues, costs and approvals for the whole company — the Board's control room.

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Hermes Agent vs Paperclip

	Hermes Agent (Day 1)	Paperclip (Day 2)
Unit	An agent (employee) — or a team of profiles	A company of agents
Orchestration	Kanban board, dependencies, swarms	Org chart, CEO delegation, projects & issues
Human control	APPROVE prompts, approval modes, blocked tasks	Board approvals for hires & strategy; in_review gates
Scheduling	Cron jobs	Routines + heartbeats
Cost control	Model choice per profile	Budgets per agent / project / company, auto-pause
Audit	Sessions, Kanban history	Immutable activity log, tool-call tracing
Best for	A personal or team agent that learns	Running recurring business functions with governance

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TOPIC 3 · DAY 2 JOURNEY

Day 2: Found → Hire → Deliver → Automate → Govern

STEP 1



Lab 6

Company, mission, budget, CEO.

STEP 2



Lab 7

Hire the team at the approval gate.

STEP 3



Lab 8

Launch project: review, revise, release.

STEP 4



Lab 9

Weekly video routine with a Board gate.

STEP 5



Lab 10

Costs, audit, scorecard, improve.

TIP Same Kopi Kaki campaign as Day 1 — compare what changes when the work is run as a company.

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TOPIC 3 · LAB 6

Lab 6 — Found Kopi Kaki Growth Studio — an AI Company You Govern

Self-host Paperclip, become the Board, set the goals and hard budgets, switch on the hiring gate and make your first agent the CEO.

The business scenario

Day 1 proved agents can do the marketing work. Kopi Kaki's owners now want it to run every week as a small 'AI growth studio' — with an org chart, budgets and an audit trail they can trust. You set up Paperclip, where you are the Board and every agent reports, ultimately, to you.

You will deliver

✓ Paperclip running at http://localhost:3100
✓ Organization 'Kopi Kaki Growth Studio' with 3 company-level goals
✓ Ava — your first agent — set as CEO on an adapter you have (Claude Code, Codex, Gemini CLI or Hermes) and passing 'Run test'
✓ 'Require board approval for new hires' switched ON
✓ Company budget USD 60/month and CEO budget USD 15/month, visible under Audit → Budgets

Agents 1 CEO agent (Ava)

Human role You are the Board: goals, budgets, the hiring gate

Time 60 min · Human-governed company of agents · labs/lab06-paperclip-found-ai-company/

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Lab 6 — How the workflow runs

Perception → reasoning → action, with you at the checkpoints.

AGENT · STEP 1

Install

npx onboard (or Docker) on localhost:3100.

HUMAN · STEP 2

Organization

Name it; create the first agent.

HUMAN · STEP 3

Adapter

Connect a model, run the test.

HUMAN · STEP 4

CEO

Make Ava the CEO; load her mandate.

HUMAN · STEP 5

Guard rails

Hiring gate ON; goals; budgets.

AGENT · STEP 6

Org chart

Board → Ava, ready to hire.

Checkpoint Agents → Org chart view shows Ava (CEO) under the Board; Audit → Budgets shows both caps; the hiring gate is ON.

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Install and Start Paperclip

TERMINAL · copy from the Lab Prompts page

```
# Option A
npx paperclipai onboard --yes

# Option B (Docker)
git clone https://github.com/paperclipai/paperclip.git && cd
paperclip
export BETTER_AUTH_SECRET="$(openssl rand -hex 32)"
export PAPERCLIP_TOOL_ACTION_SIGNING_SECRET="$(openssl rand -hex
32)"
export ANTHROPIC_API_KEY="sk-ant-..."     # and/or OPENAI_API_KEY
docker compose -f docker/docker-compose.quickstart.yml up --build
```

Option A

npx paperclipai onboard --yes — Node.js 24.11+, local, no login.

Option B

Docker: set BETTER_AUTH_SECRET + signing secret, then compose up.

Open

http://localhost:3100 → name the organization, create Ava.

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Guard Rails: CEO Role and Hard Budgets

TERMINAL · copy from the Lab Prompts page

```
npx paperclipai budget company:update -C <company-id> --payload-
json '{"budgetMonthlyCents":6000}'
npx paperclipai budget agent:update <ava-id> --payload-json
'{"budgetMonthlyCents":1500}'
npx paperclipai budget overview -C <company-id>
```

Ava = CEO

`agent update … role ceo` — the UI has no Role field.

Hiring gate

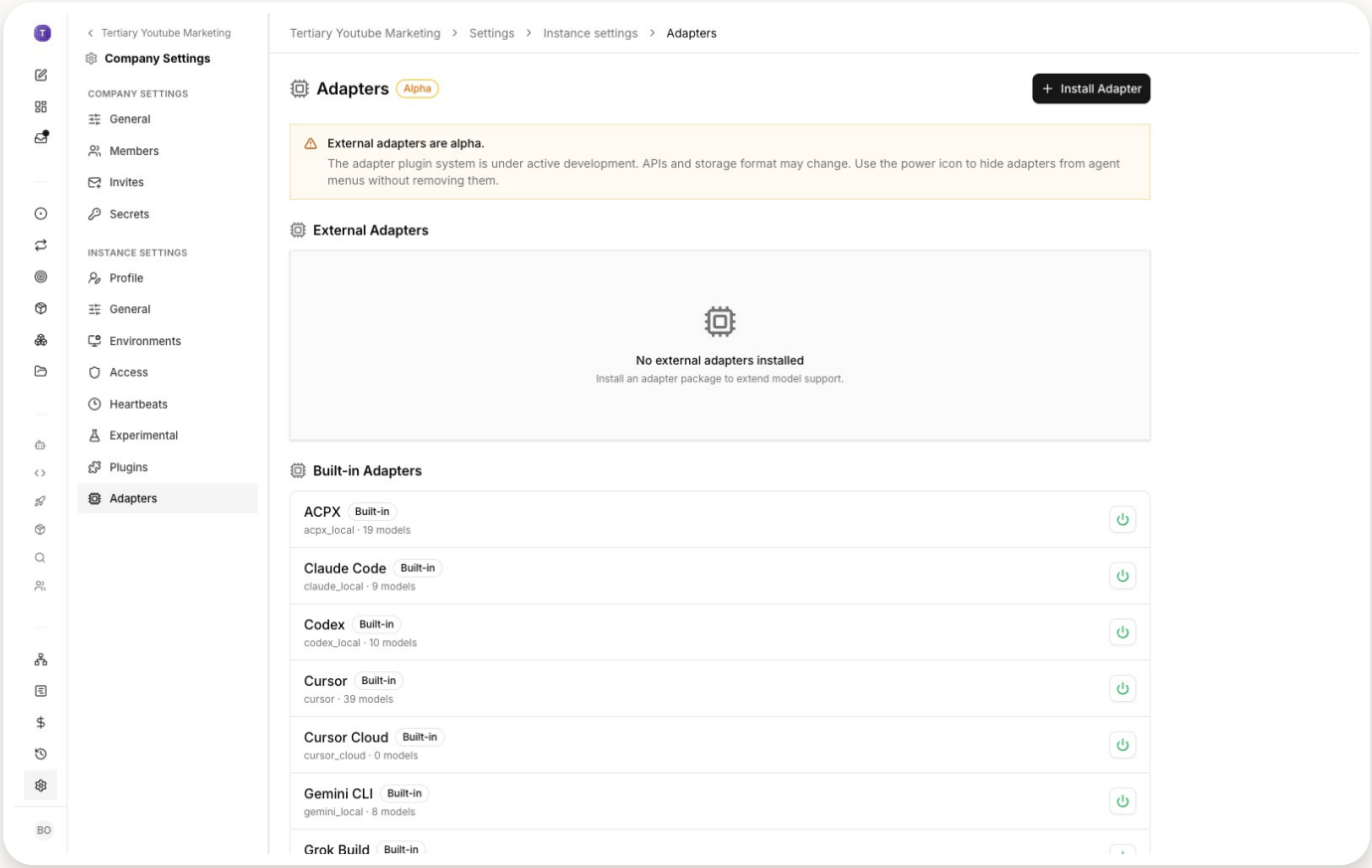
Settings → General → Hiring → require Board approval (default OFF).

Budgets

80% warning, 100% auto-pause — see Audit → Budgets.

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Adapters — the Agents' Engine



- Enable the runtime you already have — Claude Code, Codex, Gemini CLI or Hermes — and test the connection.

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Give Ava her mandate

PROMPT · copy from the Lab Prompts page

```
You are Ava, CEO of Kopi Kaki Growth Studio, a small AI marketing
studio serving Kopi Kaki Co. (a Singapore kopi chain).
Goals: (1) 2,500 app pre-orders for the Gula Melaka Cold Brew by 30
Nov 2026, (2) one approved short video every week, (3) monthly AI
spend under USD 60 with zero compliance breaches.
How you work:
- Break goals into projects and tasks; propose hires with a mandate,
adapter, reporting line and monthly budget; never hire without Board
approval.
- Every public-facing item must pass the Compliance Auditor and then
wait for the Board. Never publish anything yourself.
- Before anything is published or any spend outside budget, file a
Board approval request.
- Use brand-guide.md, campaign-brief.md and compliance-rules.md as
the source of truth.
- Report weekly: goal status, spend vs budget, risks, decisions
needed.
```



Paste into

Paperclip → Agents → Ava → Instructions → AGENTS.md



Expect

Ava's Instructions show the mandate and the three files.



You decide

You are the Board: goals, budgets, the hiring gate

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Lab 6 — Checkpoint & Reflection

Verify

http://localhost:3100 loads Kopi Kaki Growth Studio

Verify

Ava's role is CEO and her Run test passes

Verify

'Require board approval for new hires' is ON

Concepts in action

Paperclip = the company; agents = employees; you = the Board · Goal → project → task: every task traces to a company goal

Stretch challenge

Compare your Paperclip company with the Hermes studio from Lab 4: which governance features (org chart, budgets, hiring gate, audit log) would you have had to build yourself in Hermes?

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TOPIC 3 · LAB 7

Lab 7 — Hire the Marketing Team Through the Board Approval Gate

Let the CEO design the team, then approve, push back on or reshape every hire — with budgets, a Tavily web-search secret, a shared brand skill and an independent auditor.

The business scenario

Ava needs a team to hit the goals. In a real company you would never let a manager hire without approval, and you would never let the person who writes the ads also approve them. You ask Ava for a hiring plan, approve the hires that make sense, push back on one, give the researcher live web search, share the brand-voice skill, and make sure the Compliance Auditor reports to the CEO — not to the content team it audits.

You will deliver

- ✓ A plan document from Ava (roles, mandates, adapters, budgets, reporting lines)
- ✓ 5 specialists hired through Approvals: Market Researcher, Content Strategist, Copywriter, Video Producer, Compliance Auditor
- ✓ At least one hire sent back with 'Request revision' and resubmitted
- ✓ Organization secret tavily_api_key bound to the researcher as TAVILY_API_KEY
- ✓ kopi-brand-voice skill in the company library, attached to the Copywriter and Auditor

Agents CEO + 5 specialist agents

Human role Approve, revise or reject each hire; enforce separation of duties

Time 45 min · Multi-agent organisation · labs/lab07-paperclip-hire-marketing-team/

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Lab 7 — How the workflow runs

Perception → reasoning → action, with you at the checkpoints.

AGENT · STEP 1

Plan

Ava proposes the team in a plan.

HUMAN · STEP 2

Approve

Approve, revise or reject each hire.

AGENT · STEP 3

Onboard

Specialists come online with budgets.

HUMAN · STEP 4

Secret

tavily_api_key → TAVILY_API_KEY.

HUMAN · STEP 5

Skill

Share kopi-brand-voice.

AGENT · STEP 6

Smoke test

Researcher returns cited results.

Checkpoint Five specialists in the Org chart view; the auditor reports to Ava; the researcher's smoke test returns cited Tavily results.

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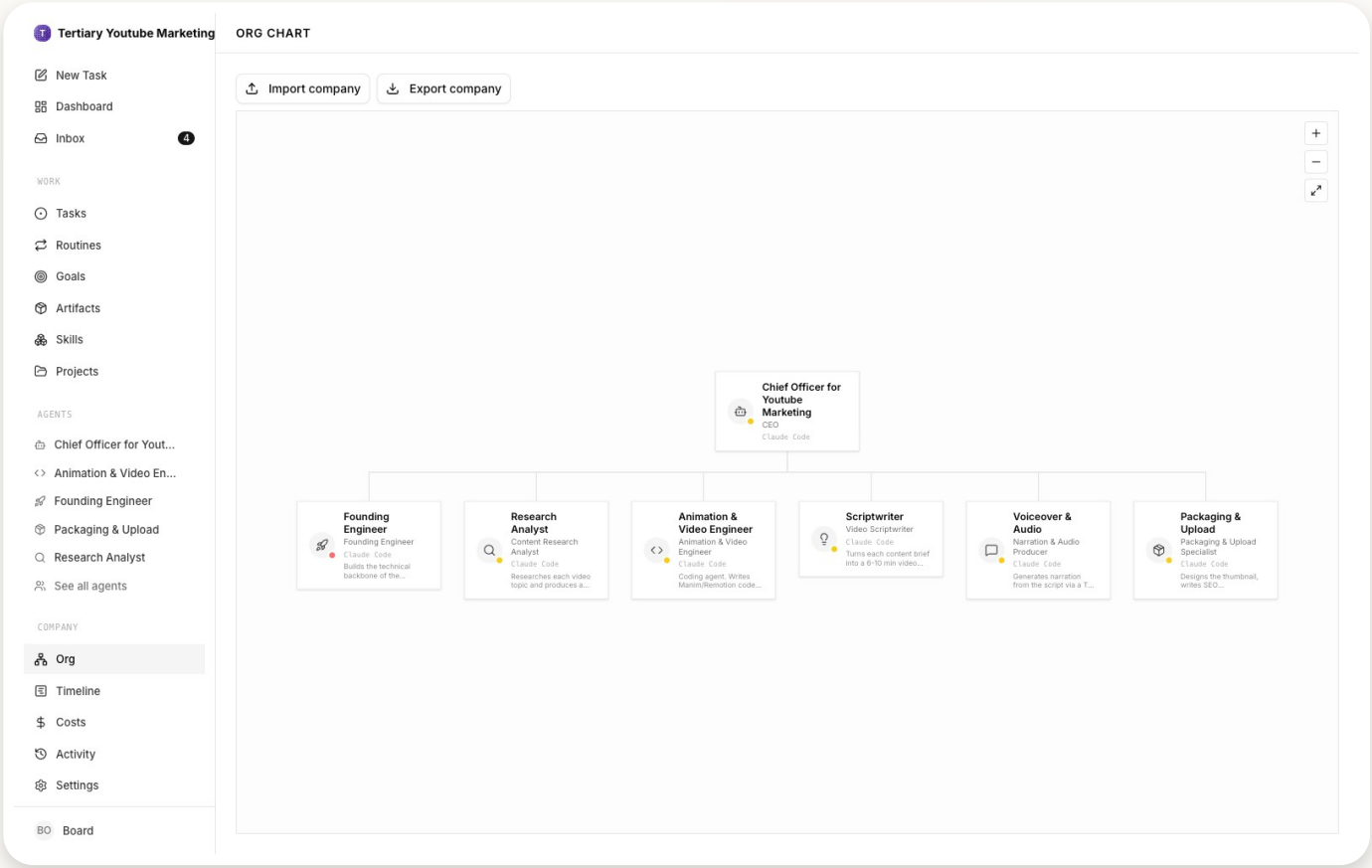
A Good Hiring Plan (reference)

Role	Reports to	Mandate	Budget / mo
Market Researcher	CEO	Live trend, competitor and audience research with Tavily; cites sources	USD 8
Content Strategist	CEO	Campaign plan, calendar and briefs	USD 8
Copywriter	Content Strategist	Captions, scripts, replies in brand voice	USD 8
Video Producer	Content Strategist	Storyboards, renders, publish package	USD 10
Compliance Auditor	CEO	Audits every public item (C1–C10, V1–V9); can block release	USD 5

TIP Separation of duties: the auditor must not report to the people whose work it audits.

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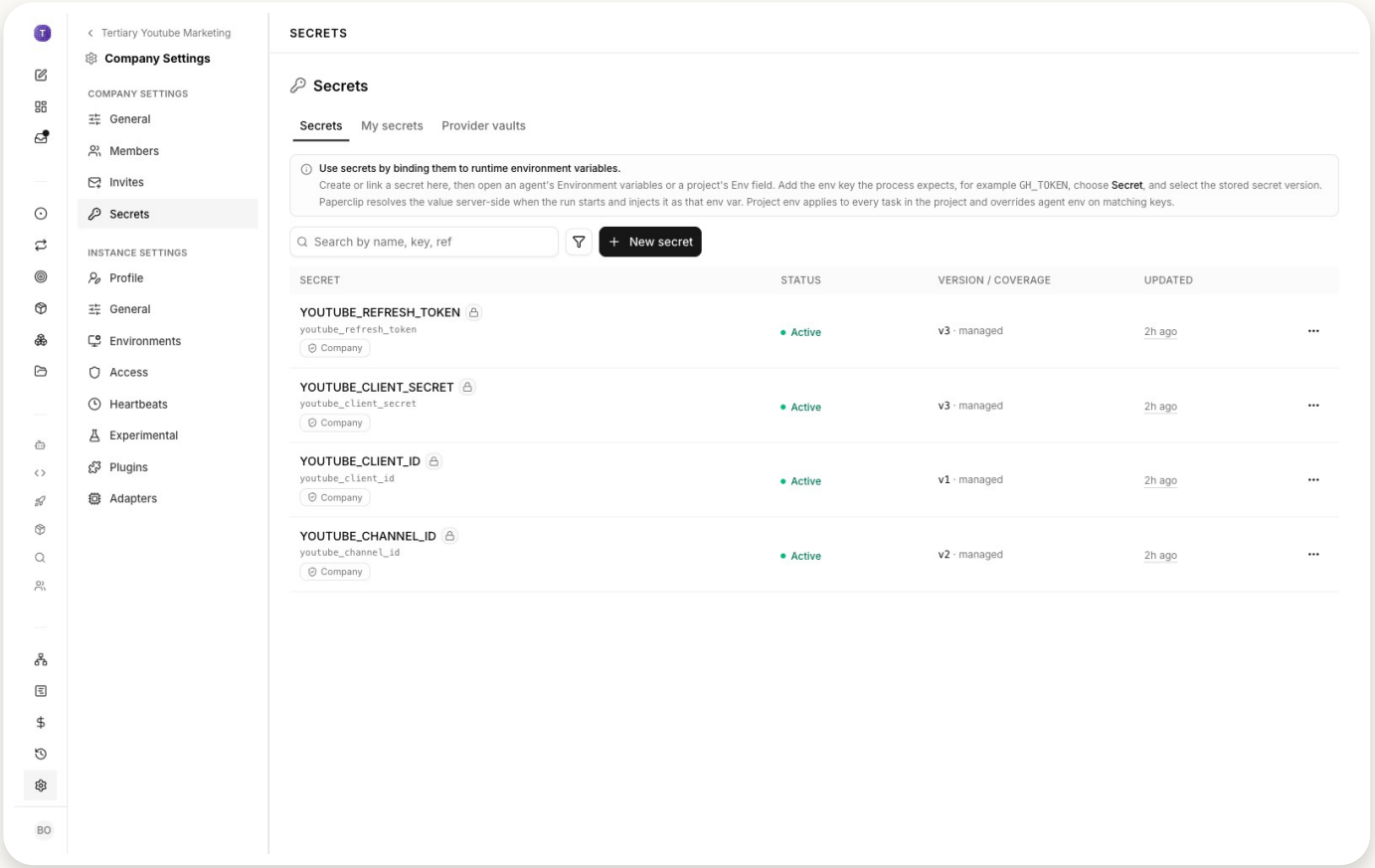
The Org Chart



- Board → CEO → specialists. Every hire passed an explicit Board decision.

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Secrets — Tools Without Exposing Keys



- Store TAVILY_API_KEY as a company Secret and bind it to the researcher's runtime — it never appears in prompts or issues.

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Ask the CEO for a hiring plan

PROMPT · copy from the Lab Prompts page

```
Propose the team to hit our three company goals.
Propose the smallest team that can research, plan, write, produce
video and audit our marketing every week. For each role give:
name, mandate (3 bullets), reports to, adapter, monthly budget in
USD, tools and skills it needs.
Rules: the Compliance Auditor must not report to anyone whose
work it audits; the total team budget must fit our USD 60/month
company budget with Ava's USD 15 included.
Save it as a plan, ask me to confirm, then submit each hire for
Board approval with its budget.
```

Paste into

Paperclip → New Task (assign to Ava)

Expect

A plan document and a confirmation card; after you confirm, hire requests appear in Approvals.

You decide

Approve, revise or reject each hire; enforce separation of duties

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Smoke-test the researcher

PROMPT · copy from the Lab Prompts page

```
Smoke test - live search.
Use Tavily to find 3 news items from the last 60 days about
coffee or cold-brew trends in Singapore or Southeast Asia. For
each: headline, date, URL and a one-line takeaway for Kopi Kaki.
Mark anything you could not verify.
```

Paste into

Paperclip → New Task (assign to Market Researcher)

Expect

Three results with URLs in the task thread.

You decide

Approve, revise or reject each hire; enforce separation of duties

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Lab 7 — Checkpoint & Reflection

Verify

5 specialists active, each with a budget

Verify

At least one hire went through Request revision

Verify

TAVILY_API_KEY bound to the researcher from an organization secret

Concepts in action

Role specialisation + reporting lines = coordination · Separation of duties: the auditor must not report to those it audits

Stretch challenge

Ask Ava to propose a sixth role — Community Manager — and decide whether the value justifies the extra budget. Record your Board decision as an approval comment.

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TOPIC 3 · LAB 8

Lab 8 — Run the #ShiokColdBrew Launch as a Project — Review, Revise, Release

Turn the campaign into seven linked tasks, let the team execute research → plan → copy → video → audit, and use an approval gate to send work back until it is good enough to release.

The business scenario

Ava's team is hired. Now it must deliver the launch campaign pack the Marketing Lead asked for on Day 1 — but as a company: tracked tasks, dependencies, owners, costs and an audit trail. Your job is the Board review: the deliverables that go public stop with you, and you either approve them or send them back with specific comments.

You will deliver

✓ Project 'ShiokColdBrew Launch' linked to goal 1, with 7 tasks and blocked-by links
✓ research-report, campaign plan + calendar, teaser captions, video script, video/storyboard + publish package
✓ Compliance audit (agent reviewer) before your approval on captions, video and release pack
✓ At least one task sent back with a comment and returned improved
✓ A retrospective with cycle time, revisions and spend

Agents CEO + 5 specialists

Human role Approve or send back every public deliverable; release

Time 55 min · Multi-agent project with Board review · labs/lab08-paperclip-launch-campaign-project/

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Lab 8 — How the workflow runs

Perception → reasoning → action, with you at the checkpoints.

AGENT · STEP 1

Backlog

Project + 7 linked tasks.

AGENT · STEP 2

Research

Cited trends and angles.

AGENT · STEP 3

Plan & write

Plan, calendar, captions, script.

AGENT · STEP 4

Produce

Video or storyboard + package.

AGENT · STEP 5

Audit

Auditor reviews C1–C10 / V1–V9.

HUMAN · STEP 6

Board review

Approve or send back; release.

Checkpoint All 7 tasks done; at least one went back from in_review with a comment and returned improved; the release pack was approved by you with a comment.

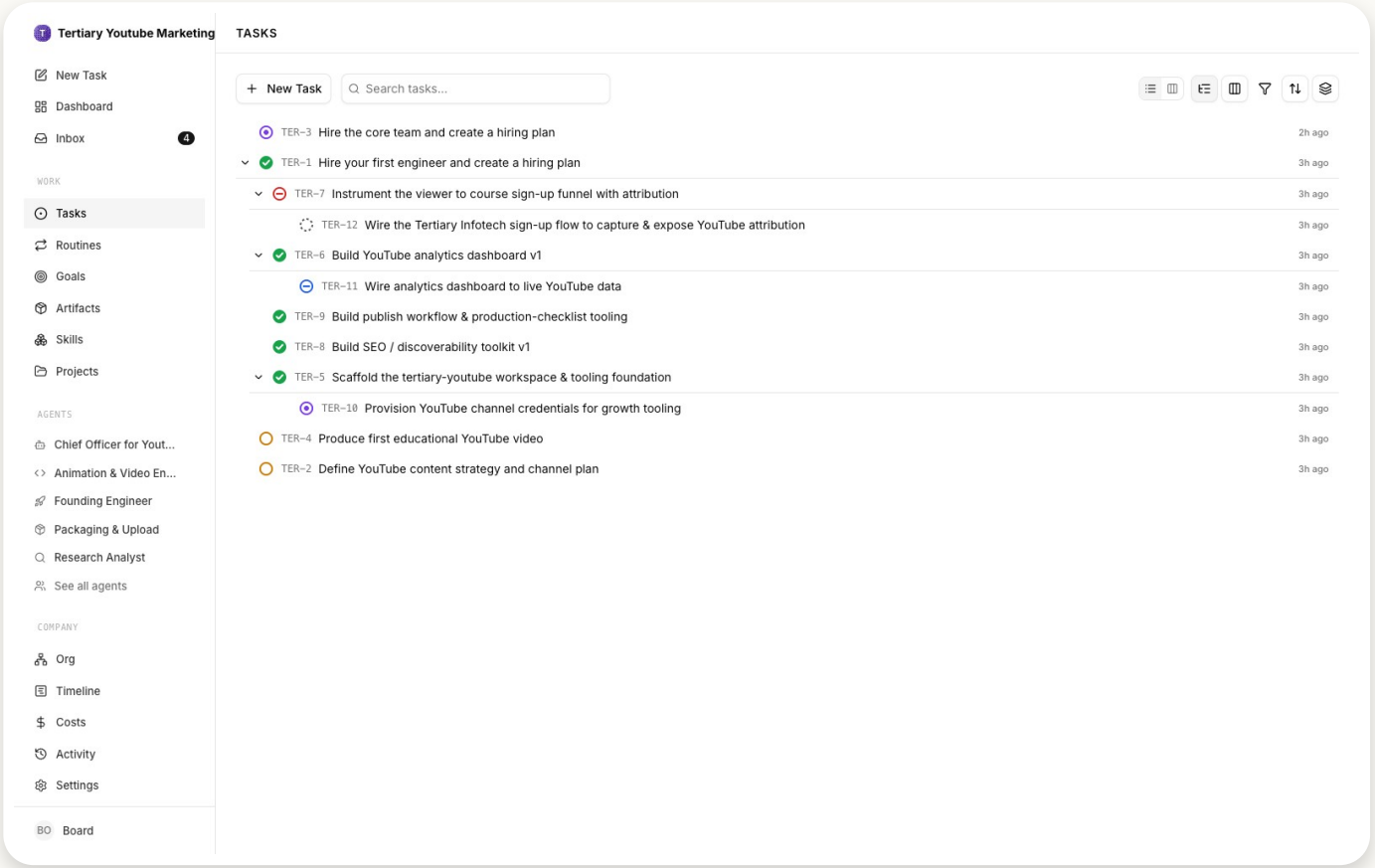
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The Launch Backlog (paperclip-backlog.csv)

#	Issue	Owner	Blocked by
1	Research SG cold-brew trends & competitor moves	Market Researcher	—
2	Campaign plan + 4-week content calendar	Content Strategist	1
3	Teaser-week captions (7 posts)	Copywriter	2
4	Script & storyboard the 25-s launch video	Video Producer	2
5	Render video + publish package	Video Producer	4
6	Compliance audit of captions and video	Compliance Auditor	3, 5
7	Board review & release pack	CEO	6

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Issues Flow to the Board



- todo → in_progress → in_review → done. in_review is your quality gate: approve, or request changes with specific comments.

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Let the CEO build the backlog

PROMPT · copy from the Lab Prompts page

```
Set up the ShiokColdBrew Launch backlog.
Create the 7 tasks in paperclip-backlog.csv inside project
"ShiokColdBrew Launch" - same titles, assignees, priorities,
blocked-by links and definitions of done (paste the definition of
done into each task). Tasks 1 and 2 must start from the Day 1
research brief and content calendar attached to the project.
Reply with the task keys.
```

Paste into

Paperclip → New Task (assign to Ava, project ShiokColdBrew Launch)

Expect

7 tasks with owners and blocked-by relationships.

You decide

Approve or send back every public deliverable; release

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Run the retrospective

PROMPT · copy from the Lab Prompts page

```
Launch project retrospective for ShiokColdBrew Launch.
Report: cycle time per task, number of Board send-backs and
auditor findings, spend per agent vs budget, and the three
biggest quality problems we caught at review. Recommend two
changes to agent instructions that would have prevented them.
```

Paste into

Paperclip → New Task (assign to Ava)

Expect

A retrospective with metrics and two instruction changes (you will apply them in Lab 10).

You decide

Approve or send back every public deliverable; release

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Lab 8 — Checkpoint & Reflection

Verify

Project linked to goal 1 with 7 linked tasks, all done

Verify

Reviewer = Compliance Auditor and Approver = you on tasks 3, 5, 7

Verify

≥ 1 send-back with comment visible in Audit → Activity

Concepts in action

Tasks + blocked-by = an ordered workflow with owners · Execution policy: agent Reviewer, then human Approver

Stretch challenge

Add a dependency the plan missed (e.g. 'Legal check of SHIOK20 terms' before release). How does one new gate change cycle time and risk?

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TOPIC 3 · LAB 9

Lab 9 — Automate the Weekly Video Pipeline with a Routine and a Board Gate

Turn research → script → production → audit → publish-ready into a routine that fires every Monday at 10:00 — and still stops for your approval.

The business scenario

Goal 2 is one approved short video every week. You do not want to create the task by hand every Monday, but you also never want a video to go out unseen. You create the 'Weekly Kopi Short' routine: the clock creates the task, Ava delegates the pipeline, and the finished package waits for a Board approval.

You will deliver

✓ Routine 'Weekly Kopi Short' for Ava — schedule trigger every week, Monday 10:00, Asia/Singapore
✓ One manual Run that produced a complete package
✓ Publish package + video/storyboard + audit in the run's task
✓ A Board Approval request that you approved (or rejected) with a comment

Agents CEO + Researcher + Copywriter + Video Producer + Auditor

Human role Approve each week's video before it is published

Time 45 min · Scheduled multi-agent workflow · labs/lab09-paperclip-weekly-video-routine/

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Lab 9 — How the workflow runs

Perception → reasoning → action, with you at the checkpoints.

AGENT · STEP 1

Trigger

Monday 10:00 creates the task.

AGENT · STEP 2

Research

Fresh angle, 2 sources.

AGENT · STEP 3

Script

20–30 s in brand voice.

AGENT · STEP 4

Produce

Video or 6-frame storyboard.

AGENT · STEP 5

Audit

PASS / FAIL with fixes.

HUMAN · STEP 6

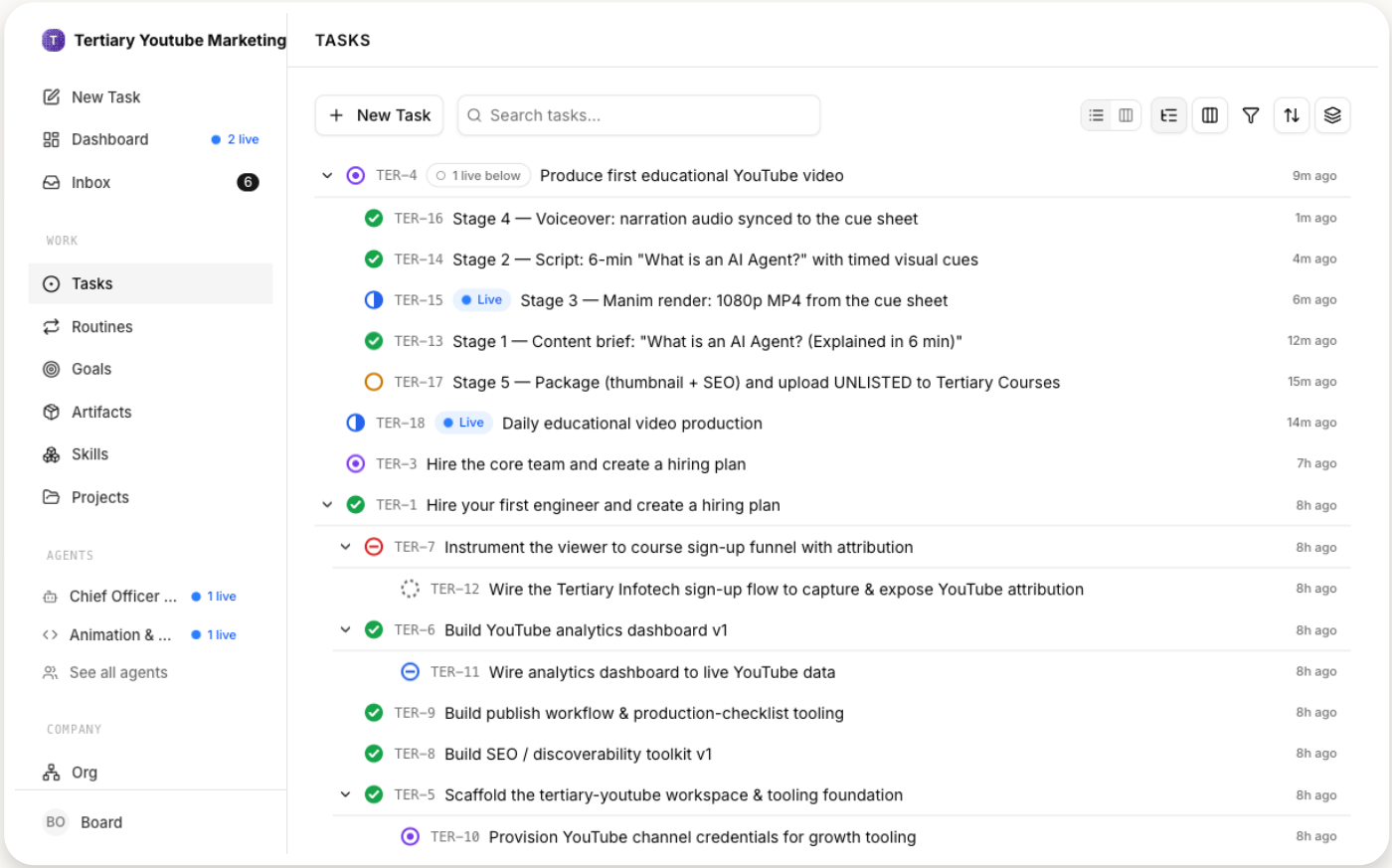
Approve

Board Approval before publishing.

Checkpoint A manual Run produced a full package and a Board Approval request; the routine shows a weekly Monday 10:00 Asia/Singapore trigger.

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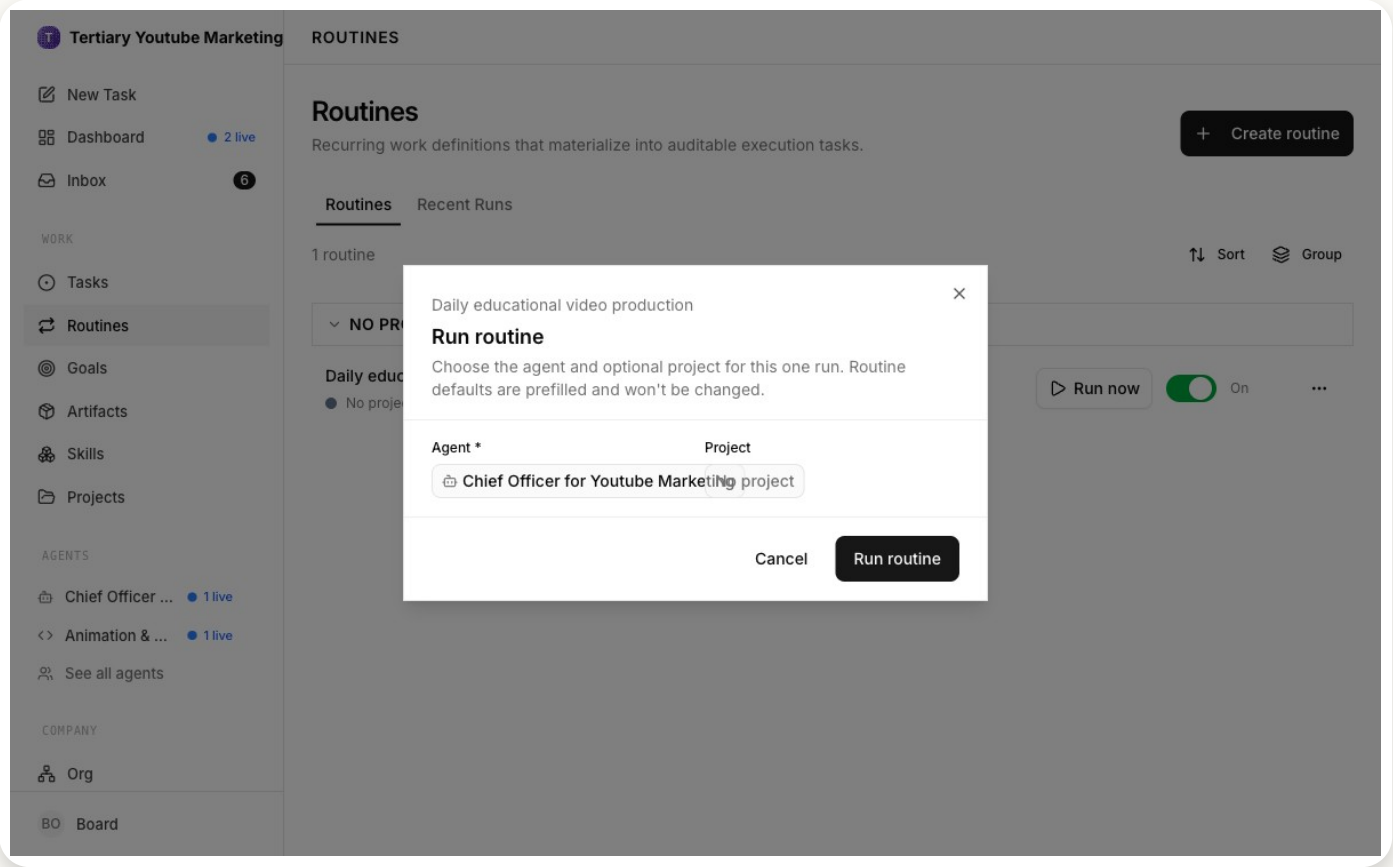
A Routine Fires the Pipeline



- Work → Routines: a schedule trigger (Monday 10:00, Asia/Singapore) generates the issue; the CEO delegates each stage.

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Run Now — Test Before You Trust



- Trigger the routine manually once and follow every stage before relying on the schedule.

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Create the routine

PROMPT · copy from the Lab Prompts page

```
Produce this week's Kopi Short (research -> script -> production
-> audit -> publish-ready).
Delegate: Market Researcher picks one fresh angle (cite 2
sources); Copywriter writes a 20-30 second script in the Kopi
Kaki voice (kopi-brand-voice); Video Producer renders the video,
or a 6-frame storyboard if rendering is unavailable; Compliance
Auditor checks compliance-rules.md (C1-C10) and video-qa-
checklist.md (V1-V9).
Deliver a publish package: title, description with AI disclosure,
3-5 hashtags incl. #ShiokColdBrew, thumbnail, captions.
Gate: before anything is published, file a Board approval request
with the package attached. Do NOT publish. Stay within each
agent's budget.
```

Paste into

Paperclip → Routines → Create routine → Instructions

Expect

The routine lands on its Triggers tab.

You decide

Approve each week's video before it is published

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Lab 9 — Checkpoint & Reflection

Verify

Routine active with a weekly Monday 10:00 Asia/Singapore trigger

Verify

Manual run created a task and completed the pipeline

Verify

The generated task shows you as Approver

Concepts in action

Routine = trigger + generated task (a cron for a company) · Heartbeats: agents wake on schedule or on assignment

Stretch challenge

Add a second trigger: a webhook that starts the pipeline when a competitor launches a product. What new risks does an event-driven trigger add, and what gate would you add?

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TOPIC 3 · LAB 10

Lab 10 — Govern, Measure & Improve Your Human–AI Workforce

Read the costs and the audit trail, score every agent, pause or re-budget one, fix the instructions that caused rework, and present a Board report on what the AI workforce delivered.

The business scenario

One month in, the owners ask the Board three questions: what did the AI workforce deliver, what did it cost, and is it safe? You use Paperclip's dashboard, Audit (Activity, Runs, Costs, Budgets) and your scorecard to answer — and you act: pause or re-budget the agent in the red, improve the instructions that caused rework, and decide what humans keep doing.

You will deliver

✓ scorecard — per-agent output, quality, cost vs budget with a RAG status
✓ One agent paused or re-budgeted, with the reason in a comment
✓ Improved instructions for the agent with the most revisions and audit findings — the Copywriter in the mock data (a new Revision you can restore)
✓ board-report — value, cost, risk, human role, next-quarter decisions — approved by you

Agents Whole company (CEO + 5 specialists)

Human role Evaluate, intervene, improve, report — and stay accountable

Time 45 min · Workforce governance · labs/lab10-paperclip-govern-measure-improve/

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Lab 10 — How the workflow runs

Perception → reasoning → action, with you at the checkpoints.

AGENT · STEP 1

Costs

Spend by agent vs budget.

AGENT · STEP 2

Audit trail

Approvals, send-backs, budget events.

AGENT · STEP 3

Scorecard

Output, quality, cost per agent.

HUMAN · STEP 4

Intervene

Pause / re-budget one agent.

HUMAN · STEP 5

Improve

Rewrite instructions that caused rework.

HUMAN · STEP 6

Report

Board report: value, cost, risk.

Checkpoint Scorecard numbers match Audit → Costs for at least two agents; one intervention and one instruction change are logged with reasons.

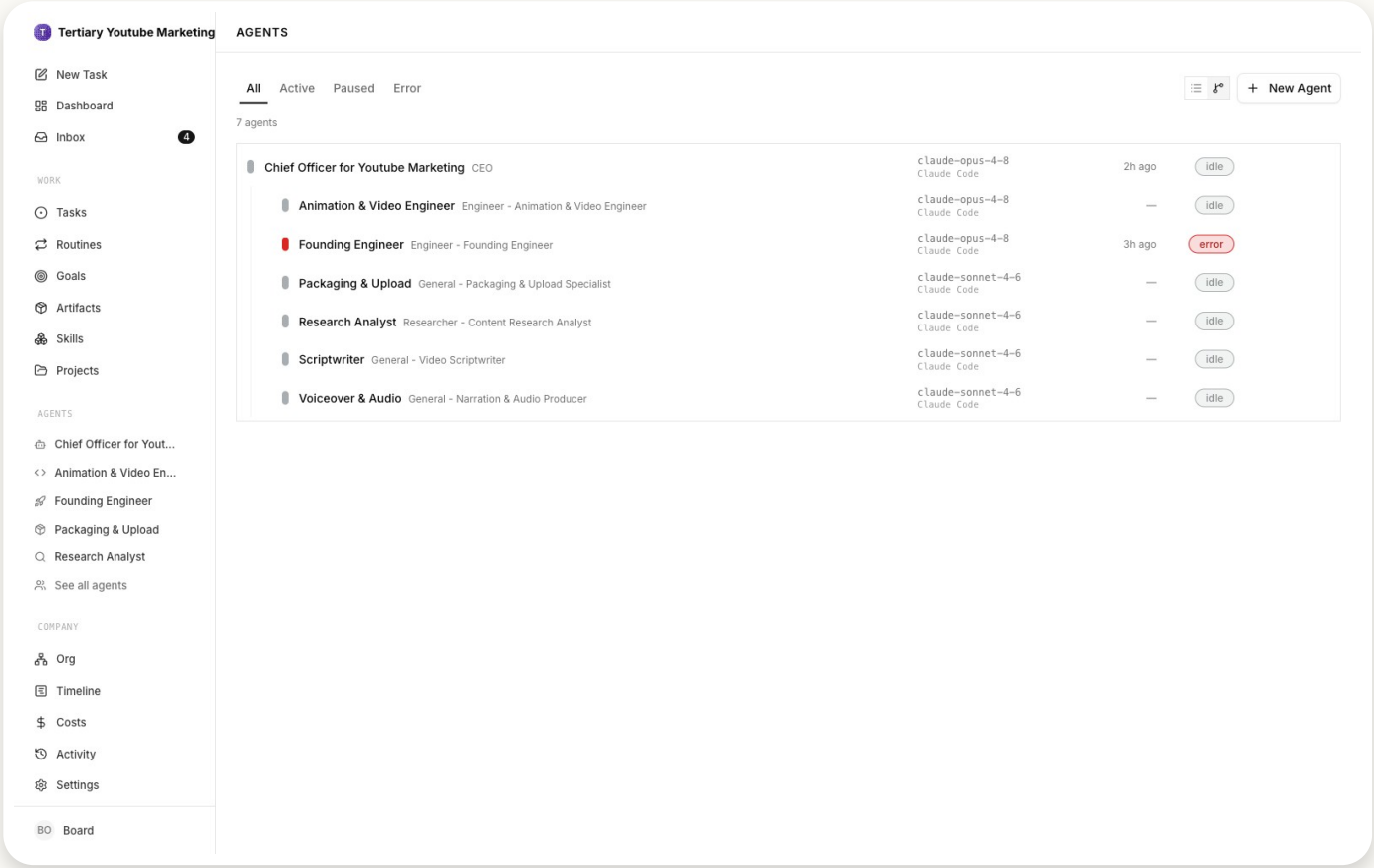
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Month-1 Agent Scorecard (agent-scorecard.csv)

Agent	Issues done	Audit FAILs	Revisions	Spend / budget (USD)	Status
Ava (CEO)	13 / 14	0	1	11.20 / 15	Green
Market Researcher	9 / 9	0	0	6.10 / 8	Green
Content Strategist	6 / 7	1	1	7.90 / 8	Amber — 99% of budget
Copywriter	18 / 22	4	3	5.30 / 8	Amber — quality
Video Producer	3 / 5	2	2	10.40 / 10	Red — over budget
Compliance Auditor	27 / 27	0	0	2.20 / 5	Green

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Agents, Budgets and Status



- Pause is reversible; terminate is not. Fix the instructions that caused rework — improve the harness.

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Build the scorecard with the CEO

PROMPT · copy from the Lab Prompts page

```
Monthly agent scorecard.
Using Paperclip's costs and activity data plus agent-
scorecard.csv, produce a scorecard with one row per agent: tasks
done, Board send-backs, audit findings, spend vs budget, cost per
completed task, and a RAG status (green/amber/red) with a one-
line reason. Show your calculations.
```

Paste into

Paperclip → New Task (assign to Ava; Approvers → Assign to me)

Expect

A scorecard you can spot-check against Audit → Costs.

You decide

Evaluate, intervene, improve, report — and stay accountable

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Improve the harness, not the blame

PROMPT · copy from the Lab Prompts page

```
Add to your rules:
- Before submitting any caption, run the kopi-brand-voice self-
check and fix every FAIL; attach the table to the task.
- Never write health or halal claims (C1, C4). Any post with
SHIOK20 must show its exact terms (C3).
- If the same rule fails twice on one task, stop and ask the
Content Strategist instead of resubmitting.
```

Paste into

Paperclip → Agents → Copywriter → Instructions (AGENTS.md)

Expect

Saved as a new Revision (Governance → Revisions → Restore is your undo). Resume the Video Producer if you paused it.

You decide

Evaluate, intervene, improve, report — and stay accountable

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Lab 10 — Checkpoint & Reflection

Verify

Scorecard values match Audit → Costs for ≥ 2 agents

Verify

A pause or budget change with a written reason

Verify

New Revision of the Copywriter's instructions

Concepts in action

Evaluate on utility, quality, cost and safety · Budget controls: 80% warning, 100% auto-pause

Stretch challenge

Decide the autonomy level (L1 Operator → L5 Observer) for each agent next quarter and justify it in one line each.

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What You Governed on Day 2

A company

Mission, goals, budget and a CEO you approved.

A team

Specialists with mandates, budgets, secrets and an independent auditor.

Delivery

A launch project and a weekly routine — all through Board review.

Governance

Costs, audit trail, scorecard, interventions and a Board report.

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WRAP-UP

Course Summary & Next Steps

1. Key takeaways
2. Practice exam
3. Assessment
4. Digital attendance

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WRAP-UP

Summary & Q&A

Topic 1

Agents loop through perception, reasoning and action; the harness and the human make them reliable.

Topic 2

Hermes took one AI employee to a supervised multi-agent team with skills, memory, cron and gates.

Topic 3

Paperclip ran the same work as a governed company — Board, budgets, routines, audit.

Your workforce

Decide per task: agent, agent team or human — and where the human approves.

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WRAP-UP

Continue Building Your Human–AI Workforce

Run 24/7

Move Hermes to a small VPS or Docker host so cron and Telegram keep running.

Codify your SOPs

Turn your team's checklists into SKILL.md skills and agent instructions.

Govern from day one

Budgets, approval gates, audit trail — follow IMDA's agentic AI framework.

Measure

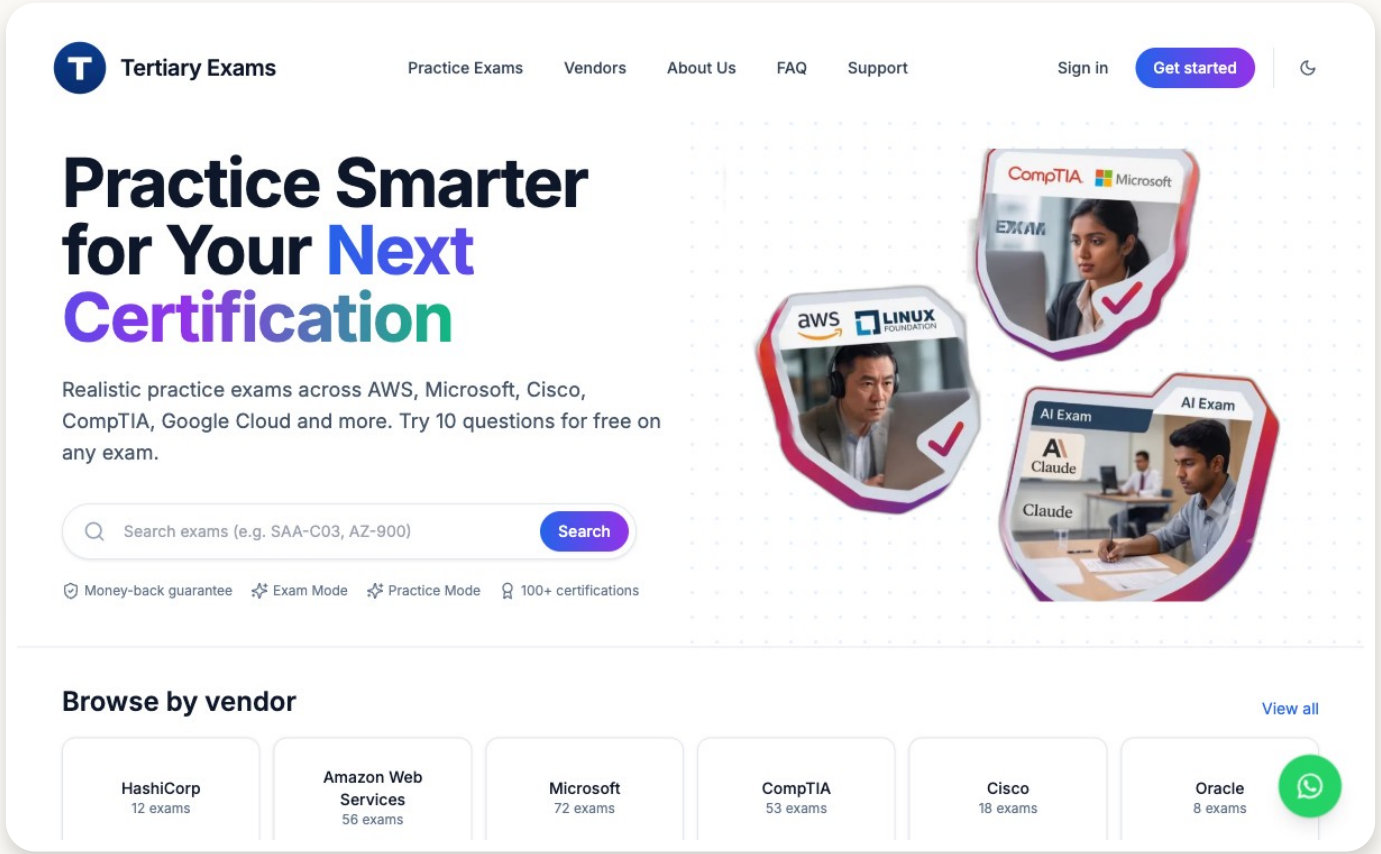
Track cycle time, revisions, cost per deliverable and quality before you scale.

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TEST YOURSELF

Practice Exam

Sharpen your readiness before the final assessment.



Open the portal

https://exams.tertiaryinfotech.com

Time yourself

Attempt it under exam conditions.

Review

Read every explanation; revisit the matching lab.

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ASSESSMENT

Final Assessment

Written Assessment (WA)

Short-Answer Questions — 1 hour, open book. Six questions on the knowledge from Topics 1–3.

Practical Performance (PP)

Three tasks — ONE screenshot each from Lab 1 (Hermes agent), Lab 4 (multi-agent studio with its human gate) and Lab 7 (Paperclip org chart), plus written analysis. 1 hour, open book.

TIP Take the Assessment digital attendance and submit your answers on the LMS at https://lms-tms.tertiaryinfotech.com/.

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ASSESSMENT

Assessment Flow

On assessment day, follow these five steps in order.

STEP 1



TRAQOM

Scan the TRAQOM QR code on the LMS and complete the survey.

STEP 2



Attendance

Take the Assessment digital attendance (SSG QR).

STEP 3



Assessment

Sit the WA (SAQ), then the PP — open book.

STEP 4



Submit

Upload your answers on the LMS.

STEP 5



Sign the record

Sign the Assessment Summary Record.

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BEFORE YOU GO

Certification, TRAQOM Survey & Digital Attendance

TRAQOM Survey (mandatory)

Complete the certification & TRAQOM survey to receive your certificate.

Scan the TRAQOM QR code on the LMS.

Digital Attendance

Take the AM, PM and Assessment digital attendance — scan the SSG QR code.

Minimum 75% attendance is required for funding.

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Thank You!

Questions? We're here to help — during and after class.

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